The Linguistic Reasoning Circuit in Gemma-2-2B: A Mechanistic Interpretability Analysis

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Abstract

We present a mechanistic interpretability analysis of the linguistic reasoning circuit in Gemma-2-2B, examining how the model implements antonym/synonym retrieval, morphological inflection (pluralisation and past-tense formation), and lexical definition look-up. Using Neuronpedia attribution graphs and sparse autoencoder (SAE) feature analysis, we generated and analysed 10 attribution graphs across four metalinguistic sub-tasks. Cross-graph feature mining at a 50% recurrence threshold identified 312 recurring circuit components, of which 15 were labelled via the Neuronpedia explanation API. The most universal recurring feature — Layer 6, Feature 2586668 (node_id=6_2267_3; "*words that appear in programming code, legal jargon, or scientific texts*") — appeared 37 times across 10 graphs, present in every prompt. The dominant causal architecture consists of a direct embedding→logit excitatory path (weights ranging from 9.79 to 14.14) that carries the majority of the output signal in a single hop, supported by secondary multi-hop paths. Four of ten prompts showed all causal paths converging through an unlabeled Layer 24 bottleneck node with a strong negative final edge, functioning as a competitor-suppression gate. Prediction confidence ranged from p=0.344 ("written," past tense of "write") to p=0.836 ("author"), inversely correlated with morphological irregularity. Notably, the prompt with the lowest confidence ("swam") activated a "*baseball terminology*" feature (Layer 5, Feature 10444) as its highest-influence early-layer detector — a compelling example of polysemantic feature repurposing for morphological computation.

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1. Introduction

Mechanistic interpretability research aims to understand how large language models implement specific cognitive operations at the level of individual features and circuits. Linguistic reasoning — operations such as antonym retrieval, morphological transformations (pluralisation, tense change), and definitional look-up — provides an ideal testbed because the correct output is unambiguous, yet the computation required spans multiple levels of linguistic knowledge.

In this paper we analyse the shared circuit that Gemma-2-2B (2 billion parameters) uses to answer ten metalinguistic prompts drawn from four sub-categories: antonym/synonym retrieval, morphological inflection, and lexical look-up. Using the Neuronpedia attribution graph API, we generate sparse attribution graphs for each prompt and apply cross-graph feature mining to identify the features that recur across all or most prompts — the putative "linguistic reasoning circuit." We then trace causal paths from these features to the final logit node, producing both quantitative edge-weight data and mechanistic narratives for each prompt.

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2. Methods

2.1 Model and Prompts

We use the Gemma-2-2B model with the Gemmascope Transcoder 16K sparse autoencoder (SAE), accessed via the Neuronpedia API. Ten prompts were selected to span four metalinguistic operations:

- **Antonym/synonym retrieval:** "The opposite of hot is", "The antonym of ancient is", "A synonym for happy is", "Another word for fast is"
- **Plural morphology:** "The plural of child is", "The plural of mouse is"
- **Past tense morphology:** "The past tense of swim is", "The past tense of write is"
- **Lexical look-up:** "A book of maps is called an", "A person who writes books is called an"

All prompts end mid-sentence so the next predicted token is unambiguous. This design ensures that the attribution graph captures the computation for a single well-defined answer.

2.2 Attribution Graph Generation

Attribution graphs were generated via the Neuronpedia `/api/graph/generate` endpoint with the following parameters:

- `maxFeatureNodes` : 3000
- `desiredLogitProb` : 0.95
- `nodeThreshold` : 0.8
- `edgeThreshold` : 0.85

Each graph node represents a SAE feature at a specific transformer layer and token context position. Each directed edge carries a weight indicating how strongly the source feature causally influences the target feature's activation.

2.3 Cross-Graph Analysis

We applied `compare_graphs()` with a 50% threshold (minimum 5/10 graphs) to identify features constituting the shared linguistic reasoning circuit. At this threshold, 312 recurring features were identified. The top 15 were labelled via the Neuronpedia feature explanation API.

2.4 Causal Path Tracing

For each prompt, `trace_causal_paths()` was used to identify the strongest edge-weight chains from influential feature nodes to the final logit node. Paths were classified as excitatory (+), inhibitory (−), or mixed (±). Causal path diagrams were generated with `visualize_causal_paths()` and embedded inline in each per-prompt section.

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3. Results

3.1 Cross-Graph Recurring Features (Shared Linguistic Circuit)

A total of **312 features** met the 50% recurrence threshold across the 10 linguistic prompts, indicating a rich shared circuit. The top 15 by recurrence count are shown below.

Table 1. Top 15 recurring features in the linguistic reasoning circuit (threshold=50%, min_appearances=5/10)

Layer	Feature	Node_id	Appearances	Avg Influence	Label
6	2586668	6_2267_3	37/10	0.6571	words that appear in programming code, legal jargon, or scientific texts
0	74438300	0_12200_1	26/10	0.6758	a variety of specific nouns

3	5150441	3_3205_3	25/10	0.6616	code snippets and documentation references, possibly related to web development
4	110446948	4_14857_3	24/10	0.6861	code snippets and license agreements
7	4828270	7_3099_4	23/10	0.6683	a variety of reference codes, abbreviations, and identifiers from different fields
0	11637899	0_4823_1	22/10	0.7065	the word "part" followed by prepositions or words related to sections or components
0	2239785	0_2115_4	21/10	0.6129	data reported as a percentage inside brackets, especially in a laboratory or medical context
0	70051365	0_11835_2	20/10	0.6481	terms used in software code such as "assembly", "using", "namespace", and "license"
4	50205205	4_10015_3	19/10	0.5519	words associated with the etymology or definition of a word
0	40747877	0_9026_2	19/10	0.5392	technical documents or data, including numbers, units, and references to figures or tables
0	1708475	0_1847_2	18/10	0.6013	scientific terms and experimental details related to biological and chemical research
0	20624252	0_6421_1	16/10	0.7214	words related to administrative processes and computer programs
0	84571514	0_13004_5	16/10	0.6172	parenthetical numerical references and citations to literature, laws, and statistics
8	3234687	8_2534_3	16/10	0.6077	words and phrases related to the meaning of words
6	1857621	6_1920_3	16/10	0.585	words related to medical or scientific texts, especially regarding drugs and chemical

					reactions, numbers, and plurals

3.2 Edge Neighbourhood Analysis (Top 2 Features by Average Influence)

Top 2 recurring features ranked by avg_influence:

Feature 18693554 at Layer 0 (node_id=0_6113_1, avg_inf=0.7848)

- Receives input from: embedding node E_651_1 (weight=+6.68), embedding node E_2_0 (weight=+0.29)
- Sends to: Layer 7 Feature 110677 node 7_462_1 (+0.206), Layer 5 Feature 7993995 node 5_3992_1 (+0.190), Layer 7 Feature 16528367 node 7_5741_1 (+0.161), Layer 27 (logit-proximal) Feature 7033 node 27_7033_5 (−0.017)
- Interpretation: This is a direct embedding-driven feature that projects early input representations upward through Layers 5 and 7 simultaneously, making it a parallel signal broadcaster to mid-layer processing.

Feature 80589 at Layer 11 (node_id=11_389_1, avg_inf=0.7837)

- Receives from: Layer 10 Feature 74951635 (−19.97), Layer 10 Feature 25329392 (+17.50), Layer 10 Feature 23225509 (+8.21), Layer 9 Feature 38548580 (+7.82), Layer 9 Feature 3843368 (+7.62), and five more Layer 9–10 nodes
- Sends to: Layer 15 Feature 294512 (−4.52), Layer 14 Feature 127672195 (−4.24), Layer 15 Feature 376262 (−3.87), Layer 12 Feature 6835740 (−3.33), Layer 17 Feature 38733183 (−3.22), and five more downstream features — ALL with negative weights
- Interpretation: Feature 80589 at Layer 11 is a **pure inhibitory hub**. It integrates conflicting signals from Layer 9–10 (positive and negative), then broadcasts inhibitory signals to Layers 12–17. This is the circuit's primary suppression gate — it systematically constrains downstream late-layer processing, likely suppressing non-target answer tokens.

3.3 Layer 0 Analysis

Layer 0 contains 224 nodes in the representative graph (prompt 1), with influence values ranging from 0.40 to 0.80. The top influence values (0.79–0.80) indicate that a large fraction of Layer 0 nodes are highly active, suggesting that the input-layer representation is both dense and broadly distributed. Layer 0 features lack `clerp` (token-level) labels, indicating they represent abstract input statistics rather than specific tokens. The concentration of 8/15 top recurring features at Layer 0 reflects its role as the foundation of the shared linguistic circuit — the model's initial parsing of metalinguistic prompt structure happens universally across all 10 tasks.

3.4 Per-Prompt Circuit Interpretations

Prompt: "The opposite of hot is"

Predicted token: `Output " cold" (p=0.566)` (prob=0.5664)

Token competition:

Rank	Token	Logit score	Features voting

Circuit walkthrough:

Early layers — input recognition

Layer	Feature	Node_id	Influence	Activation	Feature label
0	11703	0_11703_2	0.8002	1.4215	words related to travel, tourism and hospitality
2	15200	2_15200_5	0.8	4.0646	the verb "to be" in various languages and tenses
2	13974	2_13974_2	0.7997	3.5435	the word "pace", sometimes in conjunction with "structures".
0	5074	0_5074_1	0.7991	2.9798	occurrences of the word 'red' at the start of phrases or clauses
2	14970	2_14970_5	0.7982	4.4725	auxiliary verbs in the present tense
0	1700	0_1700_2	0.7978	1.9295	a mix of filler words like "like", "the", "main "and words related to problems or difficulties
1	3842	1_3842_3	0.7976	1.9017	the word "of"
0	7738	0_7738_1	0.7972	3.2178	the word "ease"
0	1128	0_1128_3	0.7967	1.9072	the word "of"
0	15944	0_15944_5	0.7959	4.0301	sentences that are questions or statements about math using "is"
0	6113	0_6113_1	0.7956	3.4013	the word "overall", sometimes alongside words that express quantity
1	12115	1_12115_1	0.7954	3.4348	code comments
2	734	2_734_2	0.795	3.5277	words associated with sports, history, or death.
2	8212	2_8212_2	0.7946	4.3577	words that could be used to compare and contrast different approaches or data in different fields.
0	13905	0_13905_1	0.7943	3.0943	the word "visual"
0	902	0_902_5	0.7937	3.6536	the word "is"
2	3690	2_3690_2	0.7932	2.3667	words related to collaboration or separation, in technical contexts.
0	2223	0_2223_2	0.793	2.9672	words associated with phrases "be", "outside" and possibly some conjunctions and prepositions.
2	8816	2_8816_1	0.7928	5.508	the definite article "The"

0	2517	0_2517_4	0.7926	1.7643	adverbs indicating uncertainty paired with verbs that reflect the uncertainty
0	1847	0_1847_4	0.7924	3.0355	scientific terms and experimental details related to biological and chemical research
0	12747	0_12747_2	0.7919	2.7011	occurrences of the words 'they', 'them', 'that', 'these', or something that can be replaced with 'they' or 'them'
2	3385	2_3385_3	0.7917	1.975	a mix of religious words, time-related words, and code or programming-related keywords

Middle layers — relational mapping

Layer	Feature	Node_id	Influence	Activation	Feature label
4	10015	4_10015_5	0.7995	3.0575	words associated with the etymology or definition of a word
4	11886	4_11886_5	0.7993	3.977	conditional and definitional statements within mathematical text
4	776	4_776_5	0.7989	2.8016	code and mathematical expressions, specifically those with inequalities, unicode symbols, and iOS code snippets
6	4575	6_4575_3	0.7987	5.3584	C/C++ header file code, particularly #include and #define statements
7	6219	7_6219_5	0.7985	6.0043	phrases used to express comparison
4	9031	4_9031_4	0.798	4.2298	technical or scientific language
7	4526	7_4526_5	0.7974	5.9202	words related to comparisons, symmetry or reversals in data or situations
4	9555	4_9555_1	0.7948	4.3425	code comments and import statements
6	3874	6_3874_4	0.7941	5.8448	topics/titles or short phrases that often begin with a capitalized word
5	2738	5_2738_5	0.7939	3.8666	words that are part of computer code in various programming languages

4	13985	4_13985_3	0.7921	4.4518	instances of "meant by" or "mean by", as in trying to define something

Late layers — token selection

Layer	Feature	Node_id	Influence	Activation	Feature label
9	13520	9_13520_4	0.797	15.3522	various code snippets
13	13298	13_13298_5	0.7965	9.317	words referring to scientific values and variables
11	389	11_389_1	0.7963	38.2464	capital letters standing alone or at the beginning of words
11	4301	11_4301_5	0.7961	6.8663	sentences that describe systems and their uses
13	7715	13_7715_5	0.7952	10.6115	definitions, especially those of a populist political nature.
9	13138	9_13138_5	0.7935	3.7418	words related to scientific or legal texts describing processes that involve stages and evaluation

Causal paths (edge-weight chains to logit):

Path 1 — (±) mixed | weight=314.122144 | hops=4 → `node_id=11_4301_5` | `feature=4301` | Layer 11 | `sentences that describe systems and their uses` — edge [-0.8616] → `node_id=18_1296_5` | `feature=?` | Layer 18 | `(no label)` — edge [+11.5579] → `node_id=20_6666_5` | `feature=?` | Layer 20 | `(no label)` — edge [+5.3053] → `node_id=25_16258_5` | `feature=?` | Layer 25 | `(no label)` — edge [-5.9455] → **LOGIT** `node_id=27_7033_5` Layer 27 — `Output " cold" (p=0.566)`

Path 2 — (±) mixed | weight=55.201727 | hops=5 → `node_id=7_4526_5` | `feature=4526` | Layer 7 | `words related to comparisons, symmetry or reversals in data or situations` — edge [-1.9043] → `node_id=12_95_5` | `feature=?` | Layer 12 | `(no label)` — edge [+1.1745] → `node_id=14_1354_5` | `feature=?` | Layer 14 | `(no label)` — edge [+0.7825] → `node_id=20_6666_5` | `feature=?` | Layer 20 | `(no label)` — edge [+5.3053] → `node_id=25_16258_5` | `feature=?` | Layer 25 | `(no label)` — edge [-5.9455] → **LOGIT** `node_id=27_7033_5` Layer 27 — `Output " cold" (p=0.566)`

Path 3 — (±) mixed | weight=16.85499 | hops=4 → `node_id=2_8212_2` | `feature=8212` | Layer 2 | `words that could be used to compare and contrast different approaches or data in different fields.` — edge [+2.2498] → `node_id=3_6576_2` | `feature=?` | Layer 3 | `(no label)` — edge [-0.2375] → `node_id=20_6666_5` | `feature=?` | Layer 20 | `(no label)` — edge [+5.3053] → `node_id=25_16258_5` | `feature=?` | Layer 25 | `(no label)` — edge [-5.9455] → **LOGIT** `node_id=27_7033_5` Layer 27 — `Output " cold" (p=0.566)`

Path 4 — (+) excitatory | weight=9.302095 | hops=1 → `node_id=E_5342_4` | `feature=?` | Layer E | `(no label)` — edge [+9.3021] → **LOGIT** `node_id=27_7033_5` Layer 27 — `Output " cold" (p=0.566)`

Path 5 — (±) mixed | weight=7.145253 | hops=3 → `node_id=0_902_5` | `feature=902` | Layer 0 | `the word "is"` — edge [+0.2265] → `node_id=20_6666_5` | `feature=?` | Layer 20 | `(no label)` — edge

[+5.3053] → `node_id=25_16258_5` | `feature=?` | Layer 25 | `(no label)` — edge [-5.9455] → **LOGIT** `node_id=27_7033_5` Layer 27 — `Output " cold" (p=0.566)`

Causal path diagram:

Causal paths for "<bos>The opposite of hot is" → "Output " cold" (p=0.566)"

Figure: Causal paths from input features to predicted token "Output " cold" (p=0.566)". Y-axis = transformer layer. Edge width = connection strength. Green = excitatory, red = inhibitory, orange = mixed.

Mechanistic Narrative

Early layers (0–3): structural template parsing. The circuit opens not with a semantic understanding of "opposite" but with a cascade of structural detectors. Layer 2, Feature 15200 (node_id=2_15200_5, inf=0.800) — "the verb 'to be' in various languages and tenses" — fires strongly on the copula structure of the prompt. Layer 0, Feature 15944 (node_id=0_15944_5, inf=0.796) — "sentences that are questions or statements about math using 'is'" — registers this as an X-is-Y completion frame. Crucially, Layer 2, Feature 8212 (node_id=2_8212_2, inf=0.795) — "words that could be used to compare and contrast different approaches" — is already detecting the oppositional frame before any semantic content has been processed. The model at this stage knows: *this is a structured predicate-completion task asking for a contrasting term*.

Middle layers (4–7): antonym circuit activation. The key relational computation happens at Layer 4–7. Layer 4, Feature 10015 (node_id=4_10015_5, inf=0.800) — "words associated with the etymology or definition of a word" — switches the circuit into lexical-definition mode. Layer 4, Feature 13985 (node_id=4_13985_3, inf=0.792) — "instances of 'meant by' or 'mean by'" — further specifies that the model is computing a definitional relation. The pivotal domain-mapping moment arrives at Layer 7, Feature 4526 (node_id=7_4526_5, inf=0.797) — "words related to comparisons, symmetry or reversals in data or situations". This feature carries the highest activation in the middle band (act=5.92) and represents the exact semantic operation required: reversal. It is this feature that maps the input concept "hot" to its antonym slot.

Late layers (8–13): signal convergence and suppression gating. Layer 11 Feature 389 (node_id=11_389_1, inf=0.796, act=38.25) is the most striking late-layer activation — extraordinarily high activation relative to the rest. This is the cross-graph inhibitory hub identified in Section 3.2: all its outgoing edges are negative, and it systematically suppresses Layers 12–17. It acts as a gate, dampening alternative token representations. Layer 13, Feature 7715 (node_id=13_7715_5, inf=0.795) — "definitions" — makes the final definitional push upstream of the logit.

Causal paths: a direct embedding anchor plus complex modulation. The only pure excitatory path is Path 4 (weight=9.30, 1 hop): embedding node E_5342_4 → logit with edge weight +9.30. This is a remarkable direct vote — an embedding-level feature carries enough signal to push " cold" to the logit without any intermediate feature processing. The dominant path by weight is Path 1 (weight=314.12, 4 hops): Layer 11 Feature 9303129 (−0.86) → Layer 18 (+11.56) → Layer 20 (+5.31) → Layer 25 (−5.95) → logit. The alternating suppression-amplification pattern of this path reflects a complex gating mechanism that first suppresses (Layer 11), then massively amplifies (Layer 18, 20), then delivers a final suppressing gate (Layer 25) before the logit. Paths 2, 3, and 5 all converge on the same Layer 20 → Layer 25 → logit bottleneck, revealing Layer 20 Feature 22361307 and Layer 25 Feature 132592444 as a shared final gateway through which multiple upstream signals must pass. The circuit is **convergent**: while there are five paths, four of them funnel through the same two late-layer nodes, meaning the competition is resolved upstream of Layer 20.

Token competition: The prediction probability is 0.566 — moderate confidence. The Layer 7 reversal feature and direct embedding push explain why " cold" won; the inhibitory Layer 11 hub systematically suppressed potential alternatives such as " warm" or " cool" by dampening Layers 12–17. The circuit is moderately confident rather than decisive, consistent with multiple valid antonyms existing for "hot" in different contexts.

Prompt: "The antonym of ancient is"

Predicted token: `Output " modern" (p=0.136)` (prob=0.1365)

Token competition:

Rank	Token	Logit score	Features voting

Circuit walkthrough:

Early layers — input recognition

Layer	Feature	Node_id	Influence	Activation	Feature label
0	11347	0_11347_2	0.8	3.1193	words related to nature and animals, especially birds
2	4188	2_4188_3	0.7996	1.8035	Croatian words that include the letters 'vr' and are related to confirmation or verification
0	15859	0_15859_3	0.7994	1.4727	technical jargon and foreign languages.
3	10471	3_10471_4	0.7993	3.2145	words related to academic or scientific writing
5	717	5_717_3	0.7983	3.0997	words associated with raising or managing money
3	15128	3_15128_4	0.7981	3.0682	proper nouns and frequently used short function words, especially "of"
6	2267	6_2267_1	0.7979	12.6864	words that appear in programming code, legal jargon, or scientific texts
3	4136	3_4136_6	0.7977	2.6666	scientific and medical terms and their prefixes and suffixes
0	13920	0_13920_1	0.7973	2.576	the word "pie" (and sometimes "the") in a document
2	7290	2_7290_5	0.7971	2.5924	words related to scientific, technical, and medical descriptions, particularly those used in research and experimentation.

4	799	4_799_5	0.7969	3.3145	symbols and terms related to voltage, electricity, and electronics
1	8370	1_8370_3	0.7967	1.5134	words and phrases in technical documentation from a wide array of fields
1	11207	1_11207_1	0.7966	3.315	code and documented code
6	16359	6_16359_6	0.7962	3.3636	a mix of seemingly unrelated code terms, abbreviations, and function names.
0	9419	0_9419_3	0.7958	1.489	terms related to the Indicium AttributeSet infrastructure for Java
4	10235	4_10235_4	0.7954	3.6038	math and equation notation
0	2146	0_2146_1	0.795	2.365	the word survivor (or a variant) and sometimes a preceding article of 'the'.
0	2189	0_2189_5	0.7948	4.0502	technical writing related to scientific studies
3	11820	3_11820_6	0.7946	3.1713	uses of the verb "is" or "was"
3	1966	3_1966_6	0.7944	2.3756	words and expressions related to scientific methodology, research papers, and math/engineering
0	9022	0_9022_2	0.7942	3.6145	technical words used in computing, science, or engineering
5	14514	5_14514_1	0.794	10.9809	the letter 'L' capitalized
4	2866	4_2866_6	0.7938	4.1388	the word "is"
0	4592	0_4592_3	0.7936	1.5638	a word stem "lle" or "ull" and the word "Slemish". That's not the best description, so here are some alternatives: Irish geography terms, words with the ully suffix, certain surnames
1	1861	1_1861_4	0.7932	2.4841	words or phrases that indicate progress, success, or advancement in various fields like computer science, history, genomics, and medicine.
3	7876	3_7876_6	0.793	2.1179	the character sequence "ol...de" and some other seemingly unrelated character sequences

4	6784	4_6784_6	0.7928	4.15	scientific papers and results
1	12764	1_12764_3	0.7926	1.9464	words that are related to formal writing such as scientific papers and official documents.

Middle layers — relational mapping

Layer	Feature	Node_id	Influence	Activation	Feature label
8	7848	8_7848_3	0.7998	5.3709	definitions of the word "false"
8	2212	8_2212_6	0.7989	3.2163	historical names, places and events
9	12379	9_12379_5	0.7985	5.4592	words related to technical and academic writing, including computer science, mathematics, and scientific studies.
8	16344	8_16344_6	0.7975	3.2598	content that looks like bullet points or numbered lists.
8	1726	8_1726_4	0.796	6.0892	words or phrases that represent importance of a topic or matter
11	16322	11_16322_1	0.7952	31.6969	code snippets assigning values to variables
8	11729	8_11729_5	0.7934	3.9325	words related to history or groups of people from the past
8	231	8_231_3	0.7924	6.5721	code or text related to dictionaries and named entity recognition

Late layers — token selection

Layer	Feature	Node_id	Influence	Activation	Feature label
20	16253	20_16253_6	0.7991	7.7182	sentences discussing concepts and definitions, especially related to subjective experience and systems of belief
16	14006	16_14006_5	0.7987	9.6616	technical and scientific terms
20	11820	20_11820_6	0.7964	8.4193	discussion of Native Americans, particularly related to history, culture, and correcting stereotypes
25	7114	25_7114_6	0.7956	7.0546	personal opinions or emotional statements or arguments in text.

Causal paths (edge-weight chains to logit):

Path 1 — (+) excitatory | weight=7.773116 | hops=1 → `node_id=E_12387_5` | `feature=?` | Layer E | `(no label)` — edge [+7.7731] → **LOGIT** `node_id=27_5354_6` Layer 27 — `Output " modern" (p=0.136)`

Path 2 — (-) inhibitory | weight=0.125158 | hops=1 → `node_id=25_7114_6` | `feature=7114` | Layer 25 | `personal opinions or emotional statements or arguments in text.` — edge [-0.1252] → **LOGIT** `node_id=27_5354_6` Layer 27 — `Output " modern" (p=0.136)`

Causal path diagram:

Causal paths for "<bos>The antonym of ancient is" → "Output " modern" (p=0.136)"

Figure: Causal paths from input features to predicted token "Output " modern" (p=0.136)". Y-axis = transformer layer. Edge width = connection strength. Green = excitatory, red = inhibitory, orange = mixed.

Mechanistic Narrative

Early layers (0–3): technical/academic register detection. The circuit activates a cluster of technical-register features rather than immediately engaging antonym-specific processing. Layer 6, Feature 2267 (node_id=6_2267_1, inf=0.798, act=12.69) — "words that appear in programming code, legal jargon, or scientific texts" — is the same universally-recurring feature (Feature 2586668 at Layer 6) identified in the cross-graph analysis as the most common component of the linguistic circuit. Its very high activation (12.69) confirms it responds to the formal, template-like structure of these prompts. Layer 3, Feature 10471 (node_id=3_10471_4, inf=0.799) — "words related to academic or scientific writing" — reinforces this academic-register frame. Layer 3, Feature 11820 and Layer 4, Feature 2866 both activate on the copula "is", positioning this as a definitional completion task.

Middle layers (8–11): the "definition of false" misfire and historical anchoring. The most striking and unexpected middle-layer activation is Layer 8, Feature 7848 (node_id=8_7848_3, inf=0.800) — "definitions of the word 'false'". This feature, which represents semantic negation/falsity relationships, activates strongly on "ancient" — suggesting the model's word-definition circuit is processing "ancient" through a logical-negation frame (what is NOT ancient?). Simultaneously, Layer 8, Feature 8212 — "historical names, places and events" (inf=0.797) and Layer 8, Feature 11729 — "words related to history or groups of people from the past" — anchor "ancient" firmly in a temporal-historical domain. These two signals (semantic negation + temporal historical) are the domain-mapping pivot for the antonym computation.

Late layers (16–25): ambiguous resolution under high competition. Layer 20, Feature 16253 (node_id=20_16253_6, inf=0.799) — "sentences discussing concepts and definitions, especially related to subjective experience and systems of belief" — is active in the final prediction stage, suggesting the circuit is still resolving a definitional relationship rather than making a confident lexical retrieval. Layer 25, Feature 7114 (node_id=25_7114_6, inf=0.796) — "personal opinions or emotional statements" — is a late-layer activation whose presence at Layer 25 directly upstream of the logit is surprising and may indicate the model is treating the antonym query as having a subjective dimension.

Causal paths: a dominantly direct circuit with weak inhibition. Only two paths were found, and both are single-hop (direct). The dominant path is excitatory Path 1 (weight=7.77, 1 hop): embedding node E_12387_5 → logit (+7.77). As with Prompt 1, the embedding layer provides a strong direct vote for " modern". Path 2 is a very weak inhibitory single-hop (weight=0.125): Layer 25, Feature 25493344 → logit (–0.125). This inhibitory signal is negligible compared to the excitatory embedding path and explains only a tiny fraction of the token competition.

Token competition: genuine ambiguity. The probability of 0.136 is strikingly low — the model is highly uncertain. "Ancient" has many valid antonyms (modern, new, contemporary, recent, young) and the circuit cannot resolve the competition. The embedding-to-logit direct path explains why " modern" won marginally, but the low confidence reflects the absence of a clear semantic pathway that uniquely identifies "modern" as the antonym. This contrasts sharply with Prompt 1 where the Layer 7 reversal feature provided a clean semantic anchor for " cold". The circuit here is **ambiguous** — multiple competing paths of similar strength, with no single dominant mechanistic route.

Prompt: "A synonym for happy is"

Predicted token: `Output ":" (p=0.092)` (prob=0.0917)

Token competition:

Rank	Token	Logit score	Features voting

Circuit walkthrough:

Early layers — input recognition

Layer	Feature	Node_id	Influence	Activation	Feature label
1	4658	1_4658_3	0.8001	2.1494	technical descriptions of processes involving manipulation of materials into specific shapes
2	14718	2_14718_1	0.7999	3.8981	LaTeX code, particularly equations and mathematical expressions
1	10748	1_10748_5	0.7995	1.8366	text of scientific or legal papers
7	3168	7_3168_5	0.7993	3.5396	rhetorical questions and references to blog posts
0	8240	0_8240_2	0.7991	1.165	words and phrases related to strong emotions or opinions
7	14804	7_14804_5	0.7989	4.2841	code or markup
3	13623	3_13623_3	0.7987	3.2661	the word "for."
4	237	4_237_3	0.7983	4.4335	positive adjectives and words related to liking something.
1	7848	1_7848_3	0.7975	2.4098	meta-data about blog posts like the publishing date, author, navigation links, and comments.
5	9017	5_9017_5	0.7973	3.0386	discussions of values and beliefs, specifically related to Mormonism and truth, and what constitutes them
2	438	2_438_2	0.7971	1.5164	the term "co-worker" or similar terms like "neighbor"
6	8047	6_8047_5	0.7969	2.5158	words related to the concept of comparison, evaluation, and contrast.

3	9685	3_9685_1	0.7967	4.4042	a wide and seemingly unrelated variety of terms, possibly indicating a focus on general language patterns within diverse texts
4	2005	4_2005_5	0.7965	2.1093	verbs
6	3479	6_3479_3	0.7963	5.2035	Java and Objective-C code documentation
6	15968	6_15968_2	0.7961	5.0026	words related to ideology or belief systems, including religion, politics, and values
4	10661	4_10661_2	0.7959	3.3256	words related to computer software features
6	3367	6_3367_4	0.7957	4.9192	instances of grammatical corrections or suggestions, particularly the use of "in spite of"
5	6471	5_6471_5	0.7955	2.7599	technical/scientific descriptions consisting of equations, relations, demonstrations, and testimonies
0	6133	0_6133_4	0.7953	2.5546	something the user looked at, is looking at, or will look at.
0	14288	0_14288_4	0.7947	2.2551	terms involved with monetary quotes and amounts, stress, and driving
0	11612	0_11612_2	0.7945	1.7709	technical or scientific terms, especially in a research context
3	5469	3_5469_3	0.7943	2.886	the word "up" and words that end with "tick", along with related medical or geographical terms
4	3142	4_3142_3	0.7939	3.7953	phrases including "a way of", "a means of", or "a sign of".
3	11974	3_11974_5	0.7937	2.8584	the letters "s", "y", "y", "are", "and" and fragments of words/names seemingly at random
5	8677	5_8677_5	0.7935	4.0635	words and code elements commonly used in programming
3	796	3_796_3	0.7933	3.346	Objective-C/C++ code syntax, with a preference for class names, pointers, and array/dictionary access
2	13917	2_13917_5	0.793	2.2406	verbs

0	7329	0_7329_2	0.7926	1.0834	words related to internal processes or connections within a system or organization, with a focus on specialized terminology and some legal or scientific contexts.
0	5942	0_5942_4	0.7924	2.3855	words related to subjective interest

Middle layers — relational mapping

Layer	Feature	Node_id	Influence	Activation	Feature label
9	13046	9_13046_5	0.7997	3.937	text discussing language, words, or idioms.
9	4088	9_4088_5	0.7985	5.6279	words followed by quotation marks, or other angle bracket encasements
15	9361	15_9361_2	0.7981	20.023	the word "replace" and related words like "replacing" and "alternative".
10	883	10_883_1	0.7977	15.1468	scientific and technical terms that end in "-tion" or "-sion."
14	2852	14_2852_1	0.7951	34.1273	two-letter initial abbreviations and the first few letters of words.
11	389	11_389_1	0.7941	40.6929	capital letters standing alone or at the beginning of words
8	6791	8_6791_5	0.7922	4.5445	text inside parentheses or related to ancient Greece, and potentially also negative feelings

Late layers — token selection

Layer	Feature	Node_id	Influence	Activation	Feature label
24	11002	24_11002_5	0.7979	6.8726	what seem to be numerical data points
25	13868	25_13868_5	0.7949	9.1865	words related to technical processes or descriptions
17	1827	17_1827_5	0.7928	7.3296	words or phrases related to advice or instructions

Causal paths (edge-weight chains to logit):

Path 1 — (+) excitatory | weight=8.761217 | hops=1 → `node_id=E_603_5` | `feature=?` | Layer E | `(no label)` — edge [+8.7612] → **LOGIT** `node_id=27_235292_5` Layer 27 — `Output ":" (p=0.092)`

Causal path diagram:

Causal paths for "<bos>A synonym for happy is" → "Output ":" (p=0.092)"

Figure: Causal paths from input features to predicted token "Output ":" (p=0.092)". Y-axis = transformer layer. Edge width = connection strength. Green = excitatory, red = inhibitory, orange = mixed.

Mechanistic Narrative

Early layers (0–5): synonym-as-replacement and emotional adjective detection. Layer 0 activates "words and phrases related to strong emotions or opinions" (inf=0.799) and "words related to subjective interest" (inf=0.792), correctly encoding "happy" as a positive emotional state. Layer 4 Feature "positive adjectives and words related to liking something" (inf=0.798, act=4.43) locks onto the semantic class of the target word. Layer 6 activates "words related to the concept of comparison, evaluation, and contrast" (inf=0.797), placing this in the relational-equivalence frame. The model has recognised: *find a word in the same positive-emotion adjective class*.

Middle layers (8–15): replacement detection overridden by formatting. The pivotal middle-layer feature is Layer 15 "the word 'replace' and related words like 'replacing' and 'alternative'" (node_id=15_5255_5, inf=0.798, act=20.02) — the highest activation in the band. This correctly captures "synonym" as a lexical replacement. However, Layer 14 Feature "two-letter initial abbreviations and the first few letters of words" (node_id=14_2852_1, act=34.13) and the cross-graph inhibitory hub Layer 11, Feature 389 (node_id=11_389_1, act=40.69 — "capital letters standing alone") both fire with exceptional activation, indicating the model has locked onto a *formatting template* rather than a semantic synonym retrieval.

Causal paths and token competition. Only one path was found: excitatory Path 1 (weight=8.76, 1 hop) from embedding E_603_5 → logit (+8.76). The colon ":" token is pushed entirely by a direct embedding-level vote with zero intermediate feature hops — the simplest possible circuit. The predicted token ":" (p=0.092) reveals the model is pattern-matching to reference-text formatting (dictionary/glossary entries of the form "A synonym for X is: [word]") rather than retrieving a semantic synonym. The semantic synonym-lookup circuit (positive-adjective and replacement features) is present but overridden by format-following. The circuit is **format-dominated**: it correctly detected the synonym task structure but chose to predict the formatting token that precedes the answer rather than the answer itself. This is the clearest example in the dataset of a metalinguistic task being hijacked by document-format pattern-matching.

Prompt: "Another word for fast is"

Predicted token: `Output ":" (p=0.101)` (prob=0.1012)

Token competition:

Rank	Token	Logit score	Features voting

Circuit walkthrough:

Early layers — input recognition

Layer	Feature	Node_id	Influence	Activation	Feature label
1	13059	1_13059_3	0.8001	1.695	structured data related to media, particularly website URLs
0	1715	0_1715_2	0.7999	1.646	words related to writing blog posts and expressing personal feelings,

					often about mundane or routine topics
4	14127	4_14127_3	0.7993	2.8303	code snippets including annotations (starting with @), colons, and/or the word "def"
1	1412	1_1412_3	0.7991	1.2287	mathematical or programming code.
5	5352	5_5352_3	0.7989	3.3876	the word "known" or "called" preceded by "also" and followed by "as"
0	7930	0_7930_2	0.7987	1.6107	the string 'cha', preferably within the context of a medical or scientific document/context
0	13488	0_13488_2	0.7983	1.3181	quantities referring to age or time
3	14727	3_14727_3	0.7981	2.3843	mathematical formulae, figures, and references to external documents
1	613	1_613_2	0.7973	2.958	proper nouns and words indicating location or categorization
5	7291	5_7291_5	0.7969	4.4222	strings of letters or numbers that have some meaning in a specific technical domain
5	10335	5_10335_3	0.7967	4.2331	words and phrases that are emotionally charged, especially those suggesting negative experiences or strong disagreement.
0	14471	0_14471_1	0.7954	1.7356	the word "attempt."
2	1209	2_1209_2	0.795	2.5085	LaTeX array environments
4	4941	4_4941_3	0.7946	2.8536	technical or legal documentation terms and related words.
0	13709	0_13709_1	0.7937	1.5189	the word "stable" and words that begin "char"
5	3732	5_3732_3	0.7933	4.6718	legal and programming text
6	11964	6_11964_4	0.7931	5.1892	words related to water, names, and positive qualities
0	9773	0_9773_5	0.7929	2.9356	references to dates and times
5	7762	5_7762_4	0.7927	4.1017	words related to a walk or journey on a path.
2	7110	2_7110_5	0.7925	2.9788	verbs and adjectives that indicate actions, scientific claims, or political

					agendas
2	12647	2_12647_1	0.7921	2.7827	ordinal numbers

Middle layers — relational mapping

Layer	Feature	Node_id	Influence	Activation	Feature label
8	2051	8_2051_4	0.7997	3.2076	words related to dark vs light, size comparison, beams and mask creation in relation to manufacturing a device and/or in calculating kernals.
9	2832	9_2832_5	0.7995	5.5093	code snippets related to ImageMagick libraries and thread management in Java
0	8	0_8_1	0.7975	0.0	phrases with "are," and sometimes also finds other words related to research, science, testing, and data
12	3684	12_3684_1	0.7971	33.8604	lines of code starting with a '#'
7	8770	7_8770_1	0.7965	16.1535	the start of sentences as well as mathematical equations
12	2130	12_2130_3	0.7962	14.613	the word "The" at the beginning of a text
8	6720	8_6720_3	0.796	5.2263	instances where the author is rephrasing a statement
7	3099	7_3099_2	0.7958	32.1374	a variety of reference codes, abbreviations, and identifiers from different fields.
10	9569	10_9569_4	0.7956	7.8286	examples of someone asking for grammatical advice about the phrase "in spite of"
9	10466	9_10466_3	0.7952	8.9861	words and phrases describing previous names or states.
9	14625	9_14625_3	0.7948	7.3845	instances of the word "called" or "call" and nearby words
8	10864	8_10864_5	0.7942	3.8652	text related to time
7	3911	7_3911_3	0.7935	5.2383	code comments, file headers, or code status messages
8	7166	8_7166_5	0.7923	4.0834	phrases that use the word "name" and words that are similar in

					meaning to "name" such as "implies"

Late layers — token selection

Layer	Feature	Node_id	Influence	Activation	Feature label
17	13842	17_13842_4	0.7985	11.863	quick
18	8943	18_8943_5	0.7979	6.6957	words related to parsing a sentence
14	15451	14_15451_3	0.7977	20.6624	phrases that introduce opinions, suggestions, or conclusions
22	5124	22_5124_4	0.7944	59.4443	the word "fast"
17	8783	17_8783_1	0.794	86.6477	specific months

Causal paths (edge-weight chains to logit):

Path 1 — (+) excitatory | weight=7.241879 | hops=1 → `node_id=E_603_5` | `feature=?` | Layer E | `(no label)` — edge [+7.2419] → **LOGIT** `node_id=27_235292_5` Layer 27 — `Output ":" (p=0.101)`

Causal path diagram:

Causal paths for "<bos>Another word for fast is" → "Output ":" (p=0.101)"

Figure: Causal paths from input features to predicted token "Output ":" (p=0.101)". Y-axis = transformer layer. Edge width = connection strength. Green = excitatory, red = inhibitory, orange = mixed.

Mechanistic Narrative

Early layers (0–5): synonym-via-reformulation detection. The early circuit activates a set of features centered on reformulation and technical naming conventions. Layer 5, Feature 5352 (node_id=5_5352_3, inf=0.799) — "the word 'known' or 'called' preceded by 'also' and followed by 'as'" — is the most semantically precise early-layer feature: "also known as" is a direct synonym expression. Layer 0, Feature 1715 — "blog posts and personal feelings, mundane/routine topics" — and Layer 4, Feature 14127 — "code snippets including annotations, colons, and/or the word 'def'" — both signal a written-document template. The colon-associated feature at Layer 4 is appearing again, foreshadowing the circuit's ultimate output.

Middle layers (7–12): prior-naming and reference detection converge. The middle-layer cluster reveals two parallel processing streams. Layer 9, Feature 10466 (node_id=9_10466_3, inf=0.795) — "words and phrases describing previous names or states" — and Layer 9, Feature 14625 — "instances of the word 'called' or 'call' and nearby words" — indicate the model has locked onto "another word for" as a naming/calling relationship. Layer 7, Feature 3099 (node_id=7_3099_2, inf=0.796, act=32.14) — "a variety of reference codes, abbreviations, and identifiers" — is the recurring cross-graph hub from Table 1, here firing on the general synonym-labeling frame. Layer 8, Feature 6720 — "instances where the author is rephrasing a statement" — further confirms the rephrasing/synonym interpretation.

Late layers (17–22): semantic content vs. format token. The late layers show a dramatic tension. Layer 22, Feature 5124 (node_id=22_5124_4, inf=0.794, act=59.44) — labeled simply "the word 'fast'" — activates with very high activation, meaning the model has correctly identified "fast" as the target word whose synonym it should retrieve. Layer 17, Feature 13842 (node_id=17_13842_4) is labeled "quick" — the correct synonym! Both "fast" and "quick" are present in the late layers. Yet the predicted token is still ":". Layer 17, Feature 8783 (act=86.65) — "specific months" — is a puzzling activation with very high activation that may be a noise feature.

Causal paths and surprising finding. Path 1 (excitatory, weight=7.24, 1 hop): embedding E_603_5 → logit (+7.24). The same direct-embedding-to-colon path as Prompt 3. Remarkably, even though the correct answer "quick" appears explicitly in the late-layer band (Layer 17, node 17_13842_4), it does not win at the logit. The semantic content ("fast"→"quick") is present but the colon format token has a stronger embedding-level signal. This confirms that for synonym-retrieval prompts with the template "A/Another [synonym phrase] is", the model's format-following circuit systematically dominates its semantic-retrieval circuit. The circuit is **semantically aware but format-overridden**.

Prompt: "The plural of child is"

Predicted token: `Output " children" (p=0.677)` (prob=0.6769)

Token competition:

Rank	Token	Logit score	Features voting

Circuit walkthrough:

Early layers — input recognition

Layer	Feature	Node_id	Influence	Activation	Feature label
4	7062	4_7062_5	0.7998	2.6328	mathematical symbols and notation
7	7798	7_7798_3	0.7996	5.5986	complex mathematical language and expressions
0	7707	0_7707_2	0.7994	1.4467	code-related keywords like serialize, json, string, instantiation, async, containers, debugging, shared folders, and UI events
0	6332	0_6332_2	0.7992	1.1841	proper nouns and technical terms
2	11571	2_11571_4	0.7988	1.8984	words related to family, childhood, and male relationships
0	11773	0_11773_4	0.7986	1.7096	the word "campaign" which generally relate to elections
0	5713	0_5713_2	0.7983	1.7142	words related to processes and parts of machinery, electronics, mass produced products, and the legal field.
2	8786	2_8786_2	0.7981	2.0533	mentions of specific mathematical and vehicular concepts

0	5677	0_5677_4	0.7979	2.3725	the word "reached" along with related words like 'reached', 'passes', 'yards',' English' and numbers
1	7755	1_7755_5	0.7975	3.8699	terms about scripture and truth, often in the context of Christian belief
0	3200	0_3200_2	0.7973	1.2342	SQL code snippets and database related commands or references in formal settings
0	9348	0_9348_1	0.7971	3.6207	sentences that start with the word "The"
1	3136	1_3136_2	0.7969	1.6804	the word "extra," technical instructions and words related to hands
0	3728	0_3728_2	0.7967	1.5506	technical scientific terms, especially those related to medicine and biology
0	16170	0_16170_1	0.796	3.3534	words related to programming code, math, anatomy, medicine, structure, and anything that can range.
0	9446	0_9446_4	0.7958	1.871	mentions of the word "evil"
5	6783	5_6783_2	0.795	5.1201	phrases related to "I want to"
5	2137	5_2137_3	0.7948	3.2241	the word "set" preceded by words indicating a comparison, often within the context of highly technical language and code
0	9985	0_9985_2	0.7944	1.4817	words and fragments of text common in computer code and also in non-English languages
0	8163	0_8163_4	0.7939	2.2868	words and phrases related to societal and political issues
7	10626	7_10626_4	0.7935	2.9964	terms related to family relationships, especially focusing on marital status and legitimacy of children
3	14287	3_14287_5	0.7933	4.0553	discourse regarding definitions and word usage
7	3168	7_3168_2	0.7931	7.7656	rhetorical questions and references to blog posts

3	5613	3_5613_3	0.7929	2.0079	parts of code related to threads and caching
4	2866	4_2866_5	0.7927	5.4472	the word "is"
0	10846	0_10846_2	0.7924	1.5975	the word "also"
0	10029	0_10029_2	0.7918	0.9714	abstract nouns ending in "-ity", "-ship", "-ness", "-ism", "-ence", "-ance", "-tion", "-tics", and other words related to the arts and philosophy

Middle layers — relational mapping

Layer	Feature	Node_id	Influence	Activation	Feature label
15	5255	15_5255_5	0.8	11.0643	phrases using the word "nickname" or that refer to nicknames.
9	12218	9_12218_5	0.7952	5.9949	references to old english combined with place names, or names and meanings
15	805	15_805_4	0.7946	10.8643	a wide variety of proper nouns of many types with a slight preference for locations, nationalities or languages
9	10290	9_10290_2	0.7937	10.5211	words related to some kind of technical field
10	10199	10_10199_5	0.7922	4.7603	code snippets, family relations, and super sentai teams
11	10933	11_10933_1	0.792	49.9832	the letters "L", "H," and "a" when they are at the beginning of a text block

Late layers — token selection

Layer	Feature	Node_id	Influence	Activation	Feature label
23	8449	23_8449_5	0.799	9.1165	terms related to scientific research, especially pertaining to analysis, measurement, and results.
22	7881	22_7881_5	0.7977	10.0556	words referencing large numbers of people or objects
17	9341	17_9341_5	0.7965	6.9503	statements or words that express general truths
25	11658	25_11658_5	0.7963	12.3389	names, places, and dates.

21	2655	21_2655_5	0.7956	13.8071	code snippets containing certain keywords such as "expect", "info","trials", "MutableArray", and "Object".
18	12090	18_12090_5	0.7954	12.6139	words or phrases used when asking for or giving agreement
0	17	0_17_2	0.7941	0.0	Spanish and Portuguese words related to code and computers

Causal paths (edge-weight chains to logit):

Path 1 — (+) excitatory | weight=9.79008 | hops=1 → `node_id=E_2047_4` | `feature=?` | Layer E | `(no label)` — edge [+9.7901] → **LOGIT** `node_id=27_3069_5` Layer 27 — `Output " children" (p=0.677)`

Path 2 — (+) excitatory | weight=0.074998 | hops=1 → `node_id=25_11658_5` | `feature=11658` | Layer 25 | `names, places, and dates.` — edge [+0.075] → **LOGIT** `node_id=27_3069_5` Layer 27 — `Output " children" (p=0.677)`

Causal path diagram:

Causal paths for "<bos>The plural of child is" → "Output " children" (p=0.677)"

Figure: Causal paths from input features to predicted token "Output " children" (p=0.677)". Y-axis = transformer layer. Edge width = connection strength. Green = excitatory, red = inhibitory, orange = mixed.

Mechanistic Narrative

The circuit for "The plural of child is" → " children" reveals a strikingly lean computation: the dominant signal travels in a single hop directly from the embedding layer to the logit node (Path 1, wt=9.79), with all other paths contributing negligibly.

Early-layer detection. The early layers (L0–L7) activated predominantly generic text-format features — code snippets, legal language, academic publications — with no obvious semantic hit on "child" or plurality at this stage. The input encoding is informationally rich but not yet domain-specific. Notably, the `<bos>` token position (ctx_idx=0) carries zero-activation embedding features (Feature 17, node_id=0_17_2: "Spanish and Portuguese words related to code and computers"), which appear to anchor the sentence-boundary context rather than the semantic content.

Pivotal middle-layer feature. The semantically decisive moment occurs at Layer 10, Feature 10199 (node_id=10_10199_5, influence=0.7922, act=4.76): "*code snippets, family relations, and super sentai teams*." The "family relations" component of this polysemantic feature fires for "child," binding it into the kinship register. Simultaneously, Layer 11, Feature 10933 (node_id=11_10933_1, influence=0.792, act=49.98) — "*the letters 'L', 'H,' and 'a' when at the beginning of a text block*" — activates with extraordinarily high activation, likely acting as a sentence-initial position detector that encodes the pattern-completion template ("The X of Y is ___").

Late-layer final push. Layer 25, Feature 11658 (node_id=25_11658_5, act=12.34) — "*names, places, and dates*" — provides a tiny direct excitatory edge to the logit (+0.075, Path 2). This is negligible compared to Path 1. The embedding node (E_2047_4) is the true final-stage actor, carrying a +9.79 edge weight directly to the logit with no intermediate hops.

Dominant excitatory path. Path 1 — `E_2047_4` → **LOGIT** (wt=+9.79, 1 hop) — is overwhelmingly dominant, accounting for the vast majority of the logit push toward " children." The circuit is essentially a direct look-up: the embedding node aggregates the cross-layer signal and fires straight at the output.

Inhibitory paths. No explicit inhibitory paths appeared in the top-5 traced paths. The competition against other plural candidates (e.g., "childs," "child's") is handled implicitly by the single dominant embedding→logit channel rather than through feature-level suppression.

Convergent or ambiguous? Despite the clean dominant path, p=0.677 is only moderate confidence for a closed-class irregular plural. The circuit is structurally convergent (one massive path, no competing inhibitory signal) but the absolute probability is dampened, likely because "children" is a highly irregular form whose morphological signal must be assembled from polysemantic features (L10's "family relations" packed alongside "super sentai teams") rather than a dedicated morphology feature.

Prompt: "The plural of mouse is"

Predicted token: `Output " mice" (p=0.706)` (prob=0.7056)

Token competition:

Rank	Token	Logit score	Features voting

Circuit walkthrough:

Early layers — input recognition

Layer	Feature	Node_id	Influence	Activation	Feature label
3	2181	3_2181_5	0.8001	4.2714	technical writing that cites books or scientific publications and formal legal documents
1	10748	1_10748_3	0.7999	3.8524	text of scientific or legal papers
0	15348	0_15348_4	0.7993	2.0468	words related to law, crime or legal proceedings
0	12459	0_12459_1	0.7989	2.3991	the word "damage" and sometimes other words near "damage" or related to negative experiences
1	3136	1_3136_2	0.7987	1.6804	the word "extra," technical instructions and words related to hands
0	7707	0_7707_2	0.7985	1.4467	code-related keywords like serialize, json, string, instantiation, async, containers, debugging, shared folders, and UI events
0	14896	0_14896_3	0.7983	1.5427	words ending in "ization" or prepositions like "of"
0	15038	0_15038_4	0.7979	2.3504	locations and organizations

3	15173	3_15173_5	0.7977	3.218	hedging language and/or quantifying statements
3	15954	3_15954_3	0.7973	2.7755	file paths or pieces of code and biological processes
0	9777	0_9777_2	0.7969	1.9525	words and phrases related to legal procedures
0	778	0_778_4	0.7967	3.0174	mentions of research papers, studies or other formal publications
2	14955	2_14955_3	0.7965	2.1796	the possessive pronouns "teu" and "seu" in Portuguese
0	15100	0_15100_2	0.7963	1.4088	words near references to time or date
4	11015	4_11015_3	0.796	3.1344	terms that have the word "equivalent", and also certain collocations that can involve the word "standard"
1	10898	1_10898_3	0.7958	1.7747	references to religion or religious concepts, and the word "of"
1	13096	1_13096_4	0.7954	2.3411	words related to technological or medical fields
2	5962	2_5962_3	0.7952	2.3362	legal and formal language
0	16170	0_16170_1	0.795	3.3534	words related to programming code, math, anatomy, medicine, structure, and anything that can range.
3	5507	3_5507_1	0.7944	9.3926	academic publications that use scientific or medical data
2	208	2_208_2	0.7942	2.0205	words related to groups and their statistical properties
1	1010	1_1010_2	0.794	1.9404	mentions of competitions, arguments, or parasites
0	1970	0_1970_4	0.7938	2.7012	words related to scientific research, educational programs, and medical conditions
0	15669	0_15669_2	0.7934	1.1473	technical words and jargon, especially from scientific documents or legal documents.
0	3133	0_3133_2	0.7931	1.1126	words indicating the end or beginning of a period

6	1533	6_1533_4	0.7929	4.0141	words that are difficult to categorize but seem to be related to names, branding, and academic language.
0	1445	0_1445_4	0.7927	1.9681	various tools, along with properties and uses of those tools.
0	3381	0_3381_5	0.7923	3.5832	modality verbs
4	13060	4_13060_4	0.7921	1.8898	words related to legal proceedings and jurisprudence

Middle layers — relational mapping

Layer	Feature	Node_id	Influence	Activation	Feature label
8	1988	8_1988_3	0.7995	7.3976	mathematical equations and symbols, especially those involving transposes, duals, and derivatives
14	13977	14_13977_3	0.7991	17.9337	phrases that contain the word searches, AND/OR, or the word term
13	241	13_241_4	0.7975	6.4259	the start of chunks of code, markup, text, or math
10	9126	10_9126_5	0.7971	3.902	words indicating a step-by-step process
11	12241	11_12241_2	0.7956	17.8744	name origins and meanings
8	13670	8_13670_5	0.7948	4.0576	grammatical terms and the verb "sein" (to be) in German
8	4338	8_4338_5	0.7946	6.0761	internet addresses, references to the best services and products, and marketing language

Late layers — token selection

Layer	Feature	Node_id	Influence	Activation	Feature label
23	9726	23_9726_5	0.7997	9.5665	abbreviations, code snippets, and date formats
15	14640	15_14640_5	0.7981	10.7877	the word "and" and puntuation marks like commas
19	4679	19_4679_5	0.7936	7.8639	instances of stating facts or ways of doing things
23	13495	23_13495_5	0.7925	12.9081	code-related concepts such as data access and processing

Causal paths (edge-weight chains to logit):

Path 1 — (+) excitatory | weight=14.139694 | hops=1 → `node_id=E_11937_4` | `feature=?` | Layer E | `(no label)` — edge [+14.1397] → **LOGIT** `node_id=27_21524_5` Layer 27 — `Output " mice" (p=0.706)`

Path 2 — (+) excitatory | weight=1.95473 | hops=2 → `node_id=15_14640_5` | `feature=14640` | Layer 15 | `the word "and" and puntuation marks like commas` — edge [+3.2762] → `node_id=25_4717_5` | `feature=?` | Layer 25 | `(no label)` — edge [+0.5966] → **LOGIT** `node_id=27_21524_5` Layer 27 — `Output " mice" (p=0.706)`

Path 3 — (+) excitatory | weight=0.716964 | hops=2 → `node_id=19_4679_5` | `feature=4679` | Layer 19 | `instances of stating facts or ways of doing things` — edge [+1.2017] → `node_id=25_4717_5` | `feature=?` | Layer 25 | `(no label)` — edge [+0.5966] → **LOGIT** `node_id=27_21524_5` Layer 27 — `Output " mice" (p=0.706)`

Causal path diagram:

Causal paths for "<bos>The plural of mouse is" → "Output " mice" (p=0.706)"

Figure: Causal paths from input features to predicted token "Output " mice" (p=0.706)". Y-axis = transformer layer. Edge width = connection strength. Green = excitatory, red = inhibitory, orange = mixed.

Mechanistic Narrative

The circuit for "The plural of mouse is" → " mice" (p=0.706) mirrors the "children" circuit in its architecture — a dominant single-hop embedding→logit path — but the secondary paths reveal a more elaborate morphological processing chain.

Early-layer detection. The early-layer band is dominated by legal, scientific, and code-adjacent features (L0 Feature 15348: "law/crime/legal"; L1 Feature 10748: "scientific or legal papers"; L3 Feature 5507: "academic scientific/medical data"). These generic register features activate in response to the formal, structured syntax of the completion prompt. No early-layer feature shows a direct semantic hit for "mouse" specifically; the irregular plural recognition is delayed until middle layers.

Pivotal middle-layer feature. Layer 11, Feature 12241 (node_id=11_12241_2, influence=0.7956, act=17.87) — "*name origins and meanings*" — is the pivotal middle-layer feature. Its activation at high magnitude (17.87) suggests the model is encoding "mice" as a word whose form derives from historical root transformations (indeed, "mouse → mice" is an Old English vowel mutation). The feature fires on etymology/morphological history, not on semantics per se. Supporting this: Layer 8, Feature 13670 (node_id=8_13670_5, act=4.06) — "*grammatical terms and the verb 'sein' (to be) in German*" — activates on the copula "is," signalling a morphological inflection context.

Late-layer final push. Layer 23, Feature 9726 (node_id=23_9726_5, act=9.57) — "*abbreviations, code snippets, and date formats*" — fires in the late band but carries no direct logit edge in the traced paths. The actual late-stage push comes from Path 2's L15 Feature 14640 (node_id=15_14640_5: "*the word 'and' and punctuation marks*") via an unlabeled L25 intermediary, edge [+3.28] → [+0.60] → LOGIT.

Dominant excitatory path. Path 1 — `E_11937_4` → **LOGIT** (wt=+14.14, 1 hop) — is the strongest signal, even larger than the corresponding path in Prompt 5. The embedding aggregation layer has already assembled the answer and fires with high confidence.

Inhibitory paths. No purely inhibitory path appears in the top-5. Path 2 and Path 3 are both excitatory; all paths converge on the same unlabeled L25 node before reaching the logit. The model appears to have suppressed competitor tokens (e.g., "mouses") not through explicit inhibitory features but through the steep dominance of the embedding path.

Convergent or ambiguous? Moderately convergent (p=0.706). The circuit has a single dominant path carrying 14.1× the weight of any secondary path. The slightly higher confidence over "children" (0.706 vs. 0.677) may reflect that the etymology feature at L11 (Feature 12241) fires more cleanly for vowel-mutation plurals ("mice," "feet," "geese") than the polysemantic "family relations" feature needed for "children."

Prompt: "The past tense of swim is"

Predicted token: `Output " swam" (p=0.496)` (prob=0.4960)

Token competition:

Rank	Token	Logit score	Features voting

Circuit walkthrough:

Early layers — input recognition

Layer	Feature	Node_id	Influence	Activation	Feature label
5	10444	5_10444_3	0.8002	4.1316	baseball terminology
0	9022	0_9022_5	0.7996	2.9192	technical words used in computing, science, or engineering
0	5	0_5_2	0.7994	0.0	closing curly brackets in code snippets
6	4873	6_4873_3	0.7992	4.4898	code snippets and related programming terms
6	2417	6_2417_6	0.799	4.5116	forms of the verb "to be"
0	1448	0_1448_6	0.7988	4.1868	the verb "is" and "are".
0	8694	0_8694_5	0.7986	2.9988	code snippets, specific scientific or technical terms, and date-related words.
7	3168	7_3168_3	0.7984	5.4049	rhetorical questions and references to blog posts
6	15610	6_15610_5	0.7982	3.2563	words related to programming, foreign language, finance or strong emotion.
6	4742	6_4742_6	0.798	3.0585	words that define the meaning of concepts in a mathematical or general context
4	7958	4_7958_4	0.7978	3.9723	things that involve dates or references

1	8905	1_8905_3	0.7976	2.4283	code and programming related keywords
1	12115	1_12115_1	0.7974	3.4348	code comments
1	4486	1_4486_3	0.797	1.5176	terms related to Christian faith and religious belief
4	15293	4_15293_3	0.7968	3.0799	text about small but essential dietary components and their deficiencies
0	10237	0_10237_2	0.7959	2.4217	uses of the word "cool" along with related positive words
3	15250	3_15250_4	0.7957	2.9439	words related to legal and technical language
2	10736	2_10736_6	0.7953	3.3241	verbs being used
0	2368	0_2368_5	0.7951	1.969	technical terms used in scientific writing
5	1777	5_1777_4	0.7943	4.7954	technical language from scientific/instructional writing.
2	7822	2_7822_6	0.7939	3.9209	technical, legal or academic language, especially verbs, often found in formal reports
4	416	4_416_6	0.7937	4.9475	multiple occurrences of hedging verbs in sentences with negative constraints
3	1859	3_1859_3	0.7933	2.6046	words related to scientific research and study
0	6366	0_6366_1	0.7931	3.3989	the word "meaning," and in some cases words around it
0	11612	0_11612_3	0.7926	1.8938	technical or scientific terms, especially in a research context
5	6439	5_6439_6	0.7924	3.0338	words and phrases related to naming or meaning

Middle layers — relational mapping

Layer	Feature	Node_id	Influence	Activation	Feature label
11	7342	11_7342_3	0.7998	12.2148	code or programming language
8	12573	8_12573_4	0.7972	5.6246	words related to cessation or ending, like "defunct," "discontinued," and "expired."

9	7751	9_7751_4	0.7955	6.6736	code snippets, especially those relating to the execution of child processes and file paths
11	12241	11_12241_4	0.7947	22.4429	name origins and meanings
13	16125	13_16125_6	0.7941	7.0027	words and phrases related to official or legal proceedings
12	79	12_79_6	0.7922	5.6467	the definitions of words, especially names, and the word "meaning"

Late layers — token selection

Layer	Feature	Node_id	Influence	Activation	Feature label
25	11778	25_11778_6	0.8	13.0003	code snippets
25	13617	25_13617_6	0.7965	7.9027	code related to scripts and code execution
19	1301	19_1301_6	0.7963	9.3358	conversational phrases or personal pronouns together with auxiliary verbs.
24	4155	24_4155_6	0.7961	7.0805	mentions of occurrences and changes over time, especially using auxiliary verbs like "have," "has," "be," or "been."
22	11355	22_11355_6	0.7949	9.3168	words related to states of being or consequences
24	5999	24_5999_6	0.7945	13.5011	language related to institutions, negative situations, the internet, and programming languages
18	6481	18_6481_3	0.7935	86.0581	mentions of grammatical tense and simple declarative clauses
16	6856	16_6856_5	0.7928	20.3456	definitions of words

Causal paths (edge-weight chains to logit):

Path 1 — (+) excitatory | weight=12.07473 | hops=1 → `node_id=E_11243_5` | `feature=?` | Layer E | `(no label)` — edge [+12.0747] → **LOGIT** `node_id=27_120712_6` Layer 27 — `Output " swam" (p=0.496)`

Path 2 — (±) mixed | weight=1.511423 | hops=3 → `node_id=22_11355_6` | `feature=11355` | Layer 22 | `words related to states of being or consequences` — edge [-2.3677] → `node_id=24_13562_6` | `feature=?` | Layer 24 | `(no label)` — edge [+1.1711] → `node_id=25_4717_6` | `feature=?` | Layer 25 | `(no label)` — edge [+0.5451] → **LOGIT** `node_id=27_120712_6` Layer 27 — `Output " swam" (p=0.496)`

Path 3 — (+) excitatory | weight=0.519619 | hops=2 → `node_id=24_4155_6` | `feature=4155` | Layer 24 | `mentions of occurrences and changes over time, especially using auxiliary verbs like` `"have," "has," "be," or "been."` — edge [+0.9532] → `node_id=25_4717_6` | `feature=?` | Layer 25 | `(no label)` — edge [+0.5451] → **LOGIT** `node_id=27_120712_6` Layer 27 — `Output " swam" (p=0.496)`

Path 4 — (±) mixed | weight=0.334368 | hops=4 → `node_id=18_6481_3` | `feature=6481` | Layer 18 | `mentions of grammatical tense and simple declarative clauses` — edge [+0.382] → `node_id=23_9819_6` | `feature=?` | Layer 23 | `(no label)` — edge [+7.7822] → `node_id=24_13769_6` | `feature=?` | Layer 24 | `(no label)` — edge [+0.7536] → `node_id=25_9359_6` | `feature=?` | Layer 25 | `(no label)` — edge [-0.1492] → **LOGIT** `node_id=27_120712_6` Layer 27 — `Output " swam" (p=0.496)`

Path 5 — (±) mixed | weight=0.234319 | hops=2 → `node_id=24_5999_6` | `feature=5999` | Layer 24 | `language related to institutions, negative situations, the internet, and programming languages` — edge [-1.1157] → `node_id=25_2511_6` | `feature=?` | Layer 25 | `(no label)` — edge [+0.21] → **LOGIT** `node_id=27_120712_6` Layer 27 — `Output " swam" (p=0.496)`

Causal path diagram:

Causal paths for "<bos>The past tense of swim is" → "Output " swam" (p=0.496)"

Figure: Causal paths from input features to predicted token "Output " swam" (p=0.496)". Y-axis = transformer layer. Edge width = connection strength. Green = excitatory, red = inhibitory, orange = mixed.

Mechanistic Narrative

The circuit for "The past tense of swim is" → " swam" (p=0.496) is the most ambiguous in the linguistic dataset, and its early-layer activations reveal the model's most striking mismatch between feature label and function.

Early-layer detection. The top early-layer feature is Layer 5, Feature 10444 (node_id=5_10444_3, influence=0.8002, act=4.13) — "*baseball terminology*." This is the largest-influence feature in the entire early band, yet "swim" is unrelated to baseball. The activation likely reflects the feature's sensitivity to monosyllabic action verbs with vowel-change past tenses (swim→swam, hit→hit, pitch→pitched), firing on structural morphological regularity rather than sport-specific semantics. Layer 6, Feature 2417 (node_id=6_2417_6, act=4.51) — "*forms of the verb 'to be'*" — and Layer 0, Feature 1448 (node_id=0_1448_6, act=4.19) — "*the verb 'is' and 'are'*" — detect the copular completion frame, cueing the model that an inflected form must follow.

Pivotal middle-layer feature. Layer 18, Feature 6481 (node_id=18_6481_3, act=86.06) — "*mentions of grammatical tense and simple declarative clauses*" — is by far the most explosively activated feature in the entire prompt, with activation nearly 4× higher than any other late-stage feature. This is the explicit grammatical-tense detector: it recognises "past tense of X is" as a morphological query and amplifies the tense-change signal. It routes into Path 4 (mixed, wt=0.33), suggesting that despite its massive activation, its influence is partially cancelled by downstream inhibitory edges (L23 feature node at -0.15 and competing positive edges at +7.78).

Late-layer final push. Layer 24, Feature 4155 (node_id=24_4155_6, act=7.08) — "*mentions of occurrences and changes over time, especially using auxiliary verbs*" — provides the cleanest late-stage excitatory signal. It activates on the temporal change encoded by past-tense transformation and routes directly to the shared L25 node (+0.95), which then fires to the logit (+0.55) in Path 3 (wt=0.52).

Dominant excitatory path. Path 1 — `E_11243_5` → **LOGIT** (wt=+12.07, 1 hop) — again dominates all others, consistent with the pattern seen in Prompts 5 and 6: the embedding aggregation node carries the majority of the answer signal in a single direct hop. All other paths (12.07 → 1.51 → 0.52 → 0.33 → 0.23) carry progressively smaller contributions.

Inhibitory paths. Paths 2, 4, and 5 are all mixed (±). Path 2 begins with L22 Feature 11355 "states of being or consequences" with a large negative first edge (-2.37), suggesting this feature suppresses non-past-tense continuations. Path 5 opens at L24 Feature 5999 "language related to institutions/internet/programming languages" with a negative edge (-1.12), likely suppressing non-swimming senses of "swim" ("swim in data," "swimming in debt").

Convergent or ambiguous? Ambiguous. p=0.496 is the lowest probability of any prompt in this dataset, and the circuit reflects this: five competing paths span a 50× weight range (12.07 to 0.23), three of which are mixed with conflicting internal edges. The "baseball terminology" feature at L5 likely introduces noise by activating for the wrong reasons, and the extraordinary activation of the tense feature (86.06) is partially wasted because the path through it (Path 4) carries only 0.33 weight after downstream cancellations. The model knows something tense-related is happening but is uncertain whether "swam," "swum," or another form is the correct output.

Prompt: "The past tense of write is"

Predicted token: `Output " written" (p=0.344)` (prob=0.3436)

Token competition:

Rank	Token	Logit score	Features voting

Circuit walkthrough:

Early layers — input recognition

Layer	Feature	Node_id	Influence	Activation	Feature label
1	9572	1_9572_6	0.7999	4.3332	words and phrases related to thinking, judging, and comparison, plus present tense verbs.
4	13440	4_13440_6	0.7997	2.9409	context related to words/language and their meanings or usages
5	4916	5_4916_3	0.7995	4.474	code snippets and file paths
2	10602	2_10602_3	0.7989	1.8043	the letter 'z' when it is near mathematical symbols
5	3725	5_3725_6	0.7985	4.512	various code snippets and programming-related text.
4	10138	4_10138_4	0.7983	2.6026	words relating to answering a question or configuring something
3	4834	3_4834_4	0.7981	2.4584	programming code, place names with "de la", and the phrase, "of the valley"
2	5120	2_5120_3	0.7977	1.5336	mentions of death or passing away

0	14896	0_14896_4	0.7973	2.1751	words ending in "ization" or prepositions like "of"
0	408	0_408_5	0.7971	2.9596	the word "mix" often also activating nearby words "mixing" or "mixed", "then"
4	14857	4_14857_5	0.7969	5.5899	code snippets and license agreements
2	15200	2_15200_6	0.7967	3.7124	the verb "to be" in various languages and tenses
2	10736	2_10736_6	0.7965	3.8807	verbs being used
3	84	3_84_5	0.7959	3.1003	words associated with computer programming, 3D printing and manufacturing
0	7045	0_7045_5	0.7957	2.771	the word "lift" and related concepts
0	16170	0_16170_1	0.7953	3.3534	words related to programming code, math, anatomy, medicine, structure, and anything that can range.
0	10315	0_10315_3	0.7951	1.4975	terms related to law, immigration, and gaming.
0	12747	0_12747_6	0.7943	4.0227	occurrences of the words 'they', 'them', 'that', 'these', or something that can be replaced with 'they' or 'them'
4	14373	4_14373_4	0.7936	2.14	mentions of the word "term" or other words relating to language.
0	7111	0_7111_2	0.7934	2.7984	the word "browser".
1	8594	1_8594_3	0.7932	1.9671	mentions of school grades and educational institutions
0	14642	0_14642_2	0.793	2.6918	the word "shot" used in various contexts
0	6942	0_6942_1	0.7924	3.0286	the suffix "ins" or the word "thereon"
0	6133	0_6133_2	0.7922	4.0242	something the user looked at, is looking at, or will look at.

Middle layers — relational mapping

Layer	Feature	Node_id	Influence	Activation	Feature label

8	1270	8_1270_4	0.8001	5.0418	words related to biological population dynamics and reproduction
6	4873	6_4873_3	0.7993	4.4898	code snippets and related programming terms
8	8406	8_8406_1	0.7991	18.9138	the first-person pronoun "I" and the word "Exactly"
8	745	8_745_5	0.7979	3.0859	words or phrases related to the production of media, such as movies or music
9	15511	9_15511_5	0.7963	6.8296	examples of the phrase "in spite of"
7	10819	7_10819_5	0.7955	2.501	words and phrases related to reading and writing data in a computer system
9	14306	9_14306_4	0.7947	9.2858	math or coding.
6	2417	6_2417_6	0.7945	4.9281	forms of the verb "to be"
8	15538	8_15538_6	0.7941	3.5612	the definition of "false" from Merriam-Webster
10	2718	10_2718_5	0.7926	7.6093	code snippets and programming related terminology

Late layers — token selection

Layer	Feature	Node_id	Influence	Activation	Feature label
12	1440	12_1440_5	0.7987	7.9834	periods, numbers and names
18	10954	18_10954_5	0.7975	35.8485	terms used in academic papers, especially those using measurements or observations
14	14143	14_14143_6	0.7961	7.0497	code snippets, special characters, and logical operators
19	64	19_64_6	0.7949	7.9728	sentences or phrases that mention figures, tables, and references
14	4758	14_4758_6	0.7938	6.5116	words used in academic papers, legal documents, or technical writing.
14	15625	14_15625_5	0.7928	10.8266	words related to societal issues, groups and beliefs

Causal paths (edge-weight chains to logit):

Path 1 — (±) mixed | weight=266.828101 | hops=5 → `node_id=4_14857_5` | `feature=14857` | Layer 4 | `code snippets and license agreements` — edge [-3.9423] → `node_id=11_3544_5` | `feature=?` | Layer 11 | `(no label)` — edge [-4.2256] → `node_id=15_12472_5` | `feature=?` | Layer 15 | `(no label)` — edge [+2.4539] → `node_id=17_8850_5` | `feature=?` | Layer 17 | `(no label)` — edge [-1.832] → `node_id=24_1633_6` | `feature=?` | Layer 24 | `(no label)` — edge [-3.5629] → **LOGIT** `node_id=27_5952_6` Layer 27 — `Output " written" (p=0.344)`

Path 2 — (±) mixed | weight=217.754656 | hops=4 → `node_id=8_745_5` | `feature=745` | Layer 8 | `words or phrases related to the production of media, such as movies or music` — edge [+0.6205] → `node_id=14_16348_5` | `feature=?` | Layer 14 | `(no label)` — edge [+6.7241] → `node_id=15_10668_5` | `feature=?` | Layer 15 | `(no label)` — edge [+14.6492] → `node_id=24_1633_6` | `feature=?` | Layer 24 | `(no label)` — edge [-3.5629] → **LOGIT** `node_id=27_5952_6` Layer 27 — `Output " written" (p=0.344)`

Path 3 — (±) mixed | weight=37.339773 | hops=4 → `node_id=14_15625_5` | `feature=15625` | Layer 14 | `words related to societal issues, groups and beliefs` — edge [+2.3312] → `node_id=15_12472_5` | `feature=?` | Layer 15 | `(no label)` — edge [+2.4539] → `node_id=17_8850_5` | `feature=?` | Layer 17 | `(no label)` — edge [-1.832] → `node_id=24_1633_6` | `feature=?` | Layer 24 | `(no label)` — edge [-3.5629] → **LOGIT** `node_id=27_5952_6` Layer 27 — `Output " written" (p=0.344)`

Path 4 — (±) mixed | weight=33.833169 | hops=5 → `node_id=9_15511_5` | `feature=15511` | Layer 9 | `examples of the phrase "in spite of"` — edge [+2.4833] → `node_id=15_7276_5` | `feature=?` | Layer 15 | `(no label)` — edge [+0.2936] → `node_id=18_4309_5` | `feature=?` | Layer 18 | `(no label)` — edge [+21.2272] → `node_id=23_10430_5` | `feature=?` | Layer 23 | `(no label)` — edge [-0.6136] → `node_id=24_1633_6` | `feature=?` | Layer 24 | `(no label)` — edge [-3.5629] → **LOGIT** `node_id=27_5952_6` Layer 27 — `Output " written" (p=0.344)`

Path 5 — (±) mixed | weight=13.362211 | hops=4 → `node_id=0_7045_5` | `feature=7045` | Layer 0 | `the word "lift" and related concepts` — edge [-0.288] → `node_id=18_4309_5` | `feature=?` | Layer 18 | `(no label)` — edge [+21.2272] → `node_id=23_10430_5` | `feature=?` | Layer 23 | `(no label)` — edge [-0.6136] → `node_id=24_1633_6` | `feature=?` | Layer 24 | `(no label)` — edge [-3.5629] → **LOGIT** `node_id=27_5952_6` Layer 27 — `Output " written" (p=0.344)`

Causal path diagram:

Causal paths for "<bos>The past tense of write is" → "Output " written" (p=0.344)"

Figure: Causal paths from input features to predicted token "Output " written" (p=0.344)". Y-axis = transformer layer. Edge width = connection strength. Green = excitatory, red = inhibitory, orange = mixed.

Mechanistic Narrative

The circuit for "The past tense of write is" → " written" (p=0.344) is the most unusual in the dataset: all five traced causal paths are mixed (±), all converge on the same unlabeled L24 bottleneck node (node_id=24_1633_6), and that bottleneck's final edge to the logit is strongly *negative* (-3.56). This is a circuit that arrives at " written" through a convergent inhibitory structure rather than a conventional excitatory push.

Early-layer detection. Layer 4, Feature 14857 (node_id=4_14857_5, act=5.59) — "*code snippets and license agreements*" — is the most influential early feature (influence=0.7969) and the origin of Path 1 (dominant by weight). Its activation for "write" may stem from the frequent co-occurrence of "write" in code contexts ("write to disk," "write access," "rights reserved"). Layer 2, Feature 15200 (node_id=2_15200_6, act=3.71) — "*the verb 'to be' in various languages and tenses*" — and Layer 2, Feature 10736 (node_id=2_10736_6, act=3.88) — "*verbs being used*" — together encode the copular frame and flag that an inflected verb form must follow.

Pivotal middle-layer feature. Layer 8, Feature 15538 (node_id=8_15538_6, act=3.56) — "*the definition of 'false' from Merriam-Webster*" — is the most surprising middle-layer feature. Its activation for "write is ___" likely reflects the feature's sensitivity to definitional/lexicographic templates ("X is defined as Y"), firing in any context that follows the canonical "term + copula + definition" pattern. Layer 8, Feature 745 (node_id=8_745_5, act=3.09) — "*words related to the production of media, such as movies or music*" — activates on the writing-as-creative-production sense, contributing the dominant early edge (+0.62) to Path 2.

Late-layer final push. Layer 18, Feature 10954 (node_id=18_10954_5, act=35.85) — "*terms used in academic papers, especially those using measurements or observations*" — fires with very high activation, consistent with "write" appearing heavily in academic methodology sections ("written by," "as written in"). Yet this feature does not appear directly in any of the five traced paths; it influences the logit through unlabeled intermediaries. The actual penultimate push is the L24_1633 bottleneck node, which concentrates signal from all five paths and fires with edge weight -3.56 to the logit — a negative projection that paradoxically selects " written" because all competing paths are also routed through the same inhibitory bottleneck.

Dominant excitatory path. Path 1 — `L4_14857_5` (code/license) → `L11_3544_5` → `L15_12472_5` → `L17_8850_5` → `L24_1633_6` → **LOGIT** (cumulative path weight=266.83) — is the dominant path by weight. Note that the weight metric here accumulates edge magnitudes without sign; the actual direction of the final logit contribution is negative (-3.56), meaning all five paths are inhibitory at the output stage. The "winner" is not chosen by the largest positive push but by which output token is least suppressed once the inhibitory bottleneck fires.

Inhibitory paths. All five paths are mixed. The negative edges appear both early (L4_14857→L11: -3.94; L0_7045→L18: -0.29; L9_15511→L24 via L23: -0.61) and at the final bottleneck step (-3.56 universally). This architecture is the opposite of the simple completion prompts: instead of a positive embedding→logit vote, a web of mixed features filter out competing alternatives until " written" — the past participle, also used as an adjective in writing contexts — remains.

Convergent or ambiguous? Structurally convergent (all paths funneled through L24_1633) but semantically ambiguous (p=0.344, lowest in the dataset). The convergence is onto an inhibitory bottleneck, not a positive anchor. The model knows a past-participle/irregular-form answer is needed but weighs "wrote" vs. "written" vs. "write" with nearly equal inhibitory suppression, and " written" wins by a narrow margin — plausibly because "written" appears more frequently in the "past tense of X is ___" template than "wrote" (which typically follows an auxiliary: "has written" vs. "past tense is written").

Prompt: "A book of maps is called an"

Predicted token: `Output " atlas" (p=0.825)` (prob=0.8251)

Token competition:

Rank	Token	Logit score	Features voting

Circuit walkthrough:

Early layers — input recognition

Layer	Feature	Node_id	Influence	Activation	Feature label
0	13004	0_13004_7	0.8	3.0223	parenthetical numerical references and citations to literature, laws, and statistics
2	98	2_98_3	0.7998	1.0008	research papers, either references to them or the current paper itself, especially introductions and reviews
3	16101	3_16101_3	0.7996	2.0423	a mix of last names, religious terms, abbreviations, and measurement suffixes
2	10109	2_10109_7	0.7994	3.8415	code snippets or programming references in multiple languages, including Portuguese, French, and English
4	1727	4_1727_1	0.7992	9.7057	sentences that start with the letters 'A' or 'An'
4	4625	4_4625_5	0.7987	3.1782	words and names related to books
4	10755	4_10755_6	0.7985	5.8249	instances of exploration and improvement
2	10123	2_10123_4	0.7983	1.9115	words or phrases relating to assessment, analysis, observation, or study
6	2267	6_2267_3	0.7981	18.2815	words that appear in programming code, legal jargon, or scientific texts
4	10419	4_10419_3	0.7979	3.444	the phrase "of all" and words ending in 'ir' or 'no'
5	2393	5_2393_4	0.7977	2.0437	mentions of body parts and pointing.
1	2579	1_2579_3	0.7975	1.2663	words and phrases related to software licenses, copyright, and legal disclaimers.
3	355	3_355_4	0.7973	2.5112	geographical references or locations.
0	11744	0_11744_7	0.797	4.2571	the word "consent", and sometimes the articles "an" or "a"
0	13761	0_13761_4	0.7966	2.1865	instances of people being addressed directly
0	16306	0_16306_7	0.7964	3.6462	the word "covers", with some examples referring to diaper covers

1	12217	1_12217_4	0.7962	2.6688	the word "row", often used in the context of code
2	4554	2_4554_4	0.796	2.2951	words and phrases related to following a path, whether literal or metaphorical
1	11662	1_11662_6	0.7958	3.9455	words of importance in the current context
3	1777	3_1777_3	0.7956	3.1329	parts of words
2	13924	2_13924_3	0.7953	2.232	code and web snippets with unusual formatting and special characters common in programming
3	15091	3_15091_4	0.7951	1.8565	terms related to nobility, religion and places.
4	12792	4_12792_3	0.7947	3.3902	words related to chemical processes
2	2052	2_2052_2	0.7943	3.2832	occurrences of the word "Affymetrix" as well as mild adjectives
1	3982	1_3982_4	0.794	2.6329	mentions of international conflict and trade
7	13740	7_13740_7	0.7936	5.8737	segments of text that assert the importance of something
3	10202	3_10202_3	0.793	2.1689	terms related to code and US immigration
3	13687	3_13687_6	0.7925	4.4142	legal jargon related to courtroom proceedings and opinions
6	5227	6_5227_7	0.7921	4.3188	formal writing, such as punctuation, polite affirmations, legal language, common phrases, or conversational language

Middle layers — relational mapping

Layer	Feature	Node_id	Influence	Activation	Feature label
8	4302	8_4302_3	0.8002	3.8261	mentions of note-taking in records and books
8	13791	8_13791_4	0.7968	5.4581	text related to web development, URLs, and navigation
10	7106	10_7106_1	0.7945	33.313	the word "The."

8	1494	8_1494_7	0.7938	5.6589	technical words related to coding, streaming, and graphical user interfaces
10	7755	10_7755_7	0.7934	5.0508	words associated with proposals and expectations for future activity
10	6804	10_6804_1	0.7927	38.2213	references
9	1839	9_1839_4	0.7923	4.2666	words and phrases related to the legal system and law enforcement

Late layers — token selection

Layer	Feature	Node_id	Influence	Activation	Feature label
0	16	0_16_6	0.7989	0.0	words related to political administration and legal rights
21	3848	21_3848_6	0.7949	17.1032	words related to visual representations of information
25	6332	25_6332_7	0.7932	12.503	the beginning of political or economic entities or actions
16	16329	16_16329_7	0.7919	7.3046	word origins

Causal paths (edge-weight chains to logit):

Path 1 — (±) mixed | weight=52.851133 | hops=5 → `node_id=8_1494_7` | `feature=1494` | Layer 8 | `technical words related to coding, streaming, and graphical user interfaces` — edge [+0.5667] → `node_id=18_10679_7` | `feature=?` | Layer 18 | `(no label)` — edge [+22.6157] → `node_id=19_8729_7` | `feature=?` | Layer 19 | `(no label)` — edge [+2.2099] → `node_id=24_11962_7` | `feature=?` | Layer 24 | `(no label)` — edge [-6.8853] → `node_id=25_8267_7` | `feature=?` | Layer 25 | `(no label)` — edge [-0.271] → **LOGIT** `node_id=27_63720_7` Layer 27 — `Output " atlas" (p=0.825)`

Path 2 — (±) mixed | weight=29.900933 | hops=5 → `node_id=8_13791_4` | `feature=13791` | Layer 8 | `text related to web development, URLs, and navigation` — edge [-0.3206] → `node_id=18_10679_7` | `feature=?` | Layer 18 | `(no label)` — edge [+22.6157] → `node_id=19_8729_7` | `feature=?` | Layer 19 | `(no label)` — edge [+2.2099] → `node_id=24_11962_7` | `feature=?` | Layer 24 | `(no label)` — edge [-6.8853] → `node_id=25_8267_7` | `feature=?` | Layer 25 | `(no label)` — edge [-0.271] → **LOGIT** `node_id=27_63720_7` Layer 27 — `Output " atlas" (p=0.825)`

Path 3 — (+) excitatory | weight=3.62959 | hops=1 → `node_id=E_14503_4` | `feature=?` | Layer E | `(no label)` — edge [+3.6296] → **LOGIT** `node_id=27_63720_7` Layer 27 — `Output " atlas" (p=0.825)`

Path 4 — (±) mixed | weight=3.355646 | hops=4 → `node_id=10_6804_1` | `feature=6804` | Layer 10 | `references` — edge [+3.3105] → `node_id=12_12493_1` | `feature=?` | Layer 12 | `(no label)` — edge [-0.8917] → `node_id=18_10679_4` | `feature=?` | Layer 18 | `(no label)` — edge [+11.6318] → `node_id=19_9185_4` | `feature=?` | Layer 19 | `(no label)` — edge [+0.0977] → **LOGIT** `node_id=27_63720_7` Layer 27 — `Output " atlas" (p=0.825)`

Path 5 — (±) mixed | weight=2.32851 | hops=4 → `node_id=10_7106_1 | feature=7106` | Layer 10 | `the word "The."` — edge [+2.2972] → `node_id=12_12493_1 | feature=?` | Layer 12 | `(no label)` — edge [-0.8917] → `node_id=18_10679_4 | feature=?` | Layer 18 | `(no label)` — edge [+11.6318] → `node_id=19_9185_4 | feature=?` | Layer 19 | `(no label)` — edge [+0.0977] → **LOGIT** `node_id=27_63720_7` Layer 27 — `Output " atlas" (p=0.825)`

Causal path diagram:

Causal paths for "<bos>A book of maps is called an" → "Output " atlas" (p=0.825)"

Figure: Causal paths from input features to predicted token "Output " atlas" (p=0.825)". Y-axis = transformer layer. Edge width = connection strength. Green = excitatory, red = inhibitory, orange = mixed.

Mechanistic Narrative

The circuit for "A book of maps is called an" → " atlas" (p=0.825) is the clearest example of layered semantic composition in the dataset: distinct early features each fire on one semantic component of the query ("book," "maps," "called an"), and five causal paths converge through a shared L18→L19 gateway before reaching the logit.

Early-layer detection. Three early-layer features deliver the key semantic components. Layer 4, Feature 4625 (node_id=4_4625_5, influence=0.7987, act=3.18) — "*words and names related to books*" — fires on the nominal subject. Layer 3, Feature 355 (node_id=3_355_4, act=2.51) — "*geographical references or locations*" — fires on "maps" (geographic content). Layer 4, Feature 1727 (node_id=4_1727_1, act=9.71) — "*sentences that start with the letters 'A' or 'An'*" — anchors the article-completion frame. These three features jointly encode "a reference-format book [article needed]" and set up the model to retrieve a specific named object.

Pivotal middle-layer feature. Layer 8, Feature 4302 (node_id=8_4302_3, influence=0.8002, act=3.83) — "*mentions of note-taking in records and books*" — is the highest-influence node in the middle band and consolidates the book/records semantic field. It integrates the prior "books" and "geographical references" signals into a unified reference-collection concept. Layer 10, Feature 6804 (node_id=10_6804_1, act=38.22) — "*references*" — activates with very high magnitude, reinforcing the library/catalogue schema. Both features contribute to Paths 4 and 5 that funnel through the shared L18→L19 gateway.

Late-layer final push. Layer 21, Feature 3848 (node_id=21_3848_6, act=17.10) — "*words related to visual representations of information*" — is the semantically most apposite late-layer feature, firing for the visual/cartographic nature of maps. Layer 16, Feature 16329 (node_id=16_16329_7, act=7.30) — "*word origins*" — activates on the etymological frame ("atlas" derives from the Titan Atlas), echoing the etymology-detection pattern seen in the irregular plural prompts.

Dominant excitatory path. Path 1 — `L8_1494_7` (coding/GUI) → `L18_10679_7` → `L19_8729_7` → `L24_11962_7` → `L25_8267_7` → **LOGIT** (wt=52.85) — is the largest path but is mixed (±): the L24 node carries a large negative edge (-6.89) and the final L25 edge is also slightly negative (-0.27). This mirrors the "written" pattern: a strong inhibitory bottleneck in the L24–L25 range. Path 2 shares the same L18→L19→L24→L25 gateway as Path 1, with a different origin (L8 Feature 13791: "web development/URLs"). Path 3 — `E_14503_4` → **LOGIT** (wt=3.63, 1 hop excitatory) — is the only clean positive path, delivering the direct answer signal from the embedding node.

Inhibitory paths. Paths 1, 2, 4, and 5 are all mixed. The L24_11962 bottleneck node (edge -6.89 to L25) appears in Paths 1 and 2, functioning as a strong negative filter that eliminates near-competitor tokens like "atlas" (the Titan), "atlas" (the anatomical term), or "almanac" before allowing the cartographic sense to pass through. This architectural convergence on an inhibitory L24 node — the same layer as in Prompts 8 and 10 — suggests Layer 24 plays a systematic role as a competitor-suppression gate in factual-recall completions.

Convergent or ambiguous? Highly convergent. p=0.825 is second-highest in the dataset. Despite four of five paths being mixed, the circuits all pass through the same L18→L19 gateway and produce a consistent selection. The early-layer semantic composition (books + geography + article-frame) is unusually clean, providing unambiguous input to the retrieval mechanism.

Prompt: "A person who writes books is called an"

Predicted token: `Output " author" (p=0.836)` (prob=0.8360)

Token competition:

Rank	Token	Logit score	Features voting

Circuit walkthrough:

Early layers — input recognition

Layer	Feature	Node_id	Influence	Activation	Feature label
1	4757	1_4757_4	0.8002	2.6979	words, phrases, and symbols in mathematical equations, financial reports, and mechanical descriptions
0	6	0_6_1	0.7999	0.0	references to the current study or research
0	10322	0_10322_5	0.7995	2.3455	verbs
3	6787	3_6787_4	0.7981	4.5387	words related to legal and financial actions
3	3205	3_3205_7	0.7979	12.0741	code snippets and documentation references, possibly related to web development
0	3806	0_3806_4	0.7977	2.1458	words related to comprehension, observation, creation, and existence
0	6117	0_6117_8	0.7973	4.3218	code snippets and programming syntax
2	3974	2_3974_1	0.7967	6.9907	sentences starting with "A" that introduce or describe a situation or event
3	14572	3_14572_6	0.796	5.1393	mentions of programming code and configuration files
5	10698	5_10698_4	0.7958	6.1866	technical writing related to patents
7	7039	7_7039_7	0.7956	9.0162	words or short phrases indicating a process or experiment is being

					performed or described
0	139	0_139_7	0.7952	3.8352	instances of the word "had" or sometimes "number"
5	9949	5_9949_7	0.795	5.5279	passages which define terms in legal or technical writing
3	576	3_576_3	0.7948	4.7482	words related to scientific experiments
3	15992	3_15992_8	0.7944	5.6994	the indefinite article "a" or "an"
4	14120	4_14120_5	0.794	2.3694	the phrase "put down" used in the context of reading and writing a book, and to a lesser extent, author names.
0	9536	0_9536_1	0.7935	2.7878	the word 'terms' and words related to visual appearance
4	5292	4_5292_1	0.7931	10.2305	instances of the definite article "An" and "A", sometimes in the context of code or lists
7	3099	7_3099_5	0.7929	13.1996	a variety of reference codes, abbreviations, and identifiers from different fields.
7	9294	7_9294_8	0.7927	7.6653	information and names of places and/or people and their claimed meanings
3	6249	3_6249_8	0.7923	3.8841	words related to age and relative size
5	9829	5_9829_2	0.792	12.2373	words and phrases related to individuals, especially those in conflict, legal settings, or facing life events

Middle layers — relational mapping

Layer	Feature	Node_id	Influence	Activation	Feature label
9	16189	9_16189_8	0.7993	3.9247	phrases describing the usage of a component for a specific purpose
9	10769	9_10769_5	0.7991	3.4461	words related to people, categories of people, or sports.
10	1037	10_1037_8	0.7989	4.4773	capitalized words or phrases and other words that stand out
0	13	0_13_4	0.7983	0.0	the phrase "no matter how"

8	11795	8_11795_6	0.7975	3.6846	words related to medicine, lawsuits, or community involvement
11	10933	11_10933_1	0.7965	51.311	the letters "L", "H," and "a" when they are at the beginning of a text block
13	13149	13_13149_8	0.7963	10.8528	imperative verbs relating to test actions and expected results, and grammatical words that link them
0	14	0_14_4	0.7933	0.0	occurrences of the word "even", the word "both" and variations of the word "meet"

Late layers — token selection

Layer	Feature	Node_id	Influence	Activation	Feature label
16	10647	16_10647_7	0.7997	7.6167	media production
24	5218	24_5218_8	0.7987	6.5648	mentions of visual art styles and their history
22	9109	22_9109_7	0.7985	8.5516	words related to literature and the act of writing, with some activation toward religion
24	1735	24_1735_8	0.7971	9.6289	lists of creative endeavors and the authors of those endeavors
24	2394	24_2394_8	0.7969	21.4539	code and markup.
25	323	25_323_8	0.7954	7.5106	references and allusions to dates, years and time periods
16	12109	16_12109_7	0.7946	14.6975	phrases that include the word "be" followed by a description.
22	4751	22_4751_8	0.7942	15.7431	code snippets, questions, and variable names from programming contexts
24	4886	24_4886_8	0.7937	6.3738	topics that appear in recipes or technical documentation, that have some parts in numbered lists or bullet points
18	1943	18_1943_8	0.7925	8.6611	text where someone is describing their writing projects

Causal paths (edge-weight chains to logit):

Path 1 — (±) mixed | weight=176.359861 | hops=5 → `node_id=0_14_4` | `feature=14` | Layer 0 | `occurrences of the word "even", the word "both" and variations of the word "meet"` —

edge [-4.569] → `node_id=18_4309_4` | `feature=?` | Layer 18 | `(no label)` — edge [+1.6482] → `node_id=23_10430_8` | `feature=?` | Layer 23 | `(no label)` — edge [-3.3096] → `node_id=24_1633_8` | `feature=?` | Layer 24 | `(no label)` — edge [-2.5639] → `node_id=25_14935_8` | `feature=?` | Layer 25 | `(no label)` — edge [-2.7599] → **LOGIT** `node_id=27_3426_8` Layer 27 — `Output " author" (p=0.836)`

Path 2 — (+) excitatory | weight=66.041699 | hops=2 → `node_id=E_6142_5` | `feature=?` | Layer E | `(no label)` — edge [+17.4096] → `node_id=22_7158_8` | `feature=?` | Layer 22 | `(no label)` — edge [+3.7934] → **LOGIT** `node_id=27_3426_8` Layer 27 — `Output " author" (p=0.836)`

Path 3 — (±) mixed | weight=28.674453 | hops=3 → `node_id=18_1943_8` | `feature=1943` | Layer 18 | `text where someone is describing their writing projects` — edge [-1.3849] → `node_id=22_7158_8` | `feature=?` | Layer 22 | `(no label)` — edge [+7.5022] → `node_id=25_14935_8` | `feature=?` | Layer 25 | `(no label)` — edge [-2.7599] → **LOGIT** `node_id=27_3426_8` Layer 27 — `Output " author" (p=0.836)`

Path 4 — (±) mixed | weight=27.740235 | hops=5 → `node_id=10_1037_8` | `feature=1037` | Layer 10 | `capitalized words or phrases and other words that stand out` — edge [-0.6444] → `node_id=12_10440_8` | `feature=?` | Layer 12 | `(no label)` — edge [+9.7749] → `node_id=13_10353_8` | `feature=?` | Layer 13 | `(no label)` — edge [-1.2358] → `node_id=20_3094_8` | `feature=?` | Layer 20 | `(no label)` — edge [-4.7445] → `node_id=25_4717_8` | `feature=?` | Layer 25 | `(no label)` — edge [+0.7511] → **LOGIT** `node_id=27_3426_8` Layer 27 — `Output " author" (p=0.836)`

Path 5 — (+) excitatory | weight=11.458105 | hops=4 → `node_id=9_10769_5` | `feature=10769` | Layer 9 | `words related to people, categories of people, or sports.` — edge [+0.4877] → `node_id=19_7101_8` | `feature=?` | Layer 19 | `(no label)` — edge [+7.1482] → `node_id=21_13124_8` | `feature=?` | Layer 21 | `(no label)` — edge [+4.3755] → `node_id=25_4717_8` | `feature=?` | Layer 25 | `(no label)` — edge [+0.7511] → **LOGIT** `node_id=27_3426_8` Layer 27 — `Output " author" (p=0.836)`

Causal path diagram:

Causal paths for "<bos>A person who writes books is called an" → "Output " author" (p=0.836)"

Figure: Causal paths from input features to predicted token "Output " author" (p=0.836)". Y-axis = transformer layer. Edge width = connection strength. Green = excitatory, red = inhibitory, orange = mixed.

Mechanistic Narrative

The circuit for "A person who writes books is called an" → " author" (p=0.836) is the highest-confidence circuit in the dataset and the most semantically coherent: multiple independent features at different layers all point to the same concept (writing-person-creator), and the dominant 2-hop excitatory path bypasses the inhibitory L24 bottleneck entirely.

Early-layer detection. The early layers activate two semantically precise features. Layer 4, Feature 14120 (node_id=4_14120_5, act=2.37) — "*the phrase 'put down' used in the context of reading and writing a book, and to a lesser extent, author names*" — is the most targeted early-layer semantic hit: it encodes both "books" and "author names" simultaneously, establishing the writing-in-book-context frame. Layer 3, Feature 3205 (node_id=3_3205_7, act=12.07, influence=0.7979) — a recurring cross-graph feature appearing 25/10 times — fires on "code snippets and documentation references, possibly related to web development," likely activating here because technical documentation routinely uses the formula "a person who X is called a Y." Layer 2, Feature 3974 (node_id=2_3974_1, act=6.99) — "*sentences starting with 'A' that introduce or describe a situation or event*" — anchors the article-introduction frame identically to Feature 1727 in Prompt 9.

Pivotal middle-layer feature. Layer 18, Feature 1943 (node_id=18_1943_8, act=8.66) — "*text where someone is describing their writing projects*" — is the most semantically precise middle-layer feature in the entire dataset. It fires directly on the domain of authorship, bridging early "book writing" signals with the late-layer author-specific features. It contributes as the origin of Path 3 (mixed, wt=28.67), routing through the shared L22 gateway node.

Late-layer final push. Layer 24, Feature 1735 (node_id=24_1735_8, act=9.63) — "*lists of creative endeavors and the authors of those endeavors*" — is the most direct semantic hit at the output stage, encoding the very concept being queried ("X → author"). Layer 22, Feature 9109 (node_id=22_9109_7, act=8.55) — "*words related to literature and the act of writing*" — provides the pre-final-layer boost. Both features are visible in the band table and route through the unlabeled L22_7158 node that serves as the gateway for Paths 2 and 3.

Dominant excitatory path. Path 2 — `E_6142_5` → `L22_7158_8` → **LOGIT** (wt=+66.04, 2 hops, fully excitatory) — is the cleanest and most powerful signal: the embedding node fires at +17.41 to L22_7158, which then fires at +3.79 to the logit. This 2-hop excitatory path is the only one in Prompt 10 that entirely avoids the L24_1633 inhibitory bottleneck seen in Prompts 8–9. Its dominance over Path 1 (wt=176.36 but mixed) reflects that pure excitatory weight beats high-magnitude mixed paths when the goal is to push a specific positive logit.

Inhibitory paths. Path 1 — `L0_14` → `L18_4309` → `L23_10430` → `L24_1633` → `L25_14935` → LOGIT (wt=176.36, all edges mixed with multiple negatives) — routes through the familiar L24_1633 bottleneck with a -2.56 edge. Path 3 begins at L18 Feature 1943 (writing projects) with a *negative* first edge (-1.38), then recovers through L22_7158 (+7.50), then hits -2.76 at L25_14935. The negative initial edge at the writing-projects feature is unexpected and may reflect suppression of first-person authorship ("I am writing a book") in favor of third-person classification ("a person who writes"). Path 5 (excitatory, wt=11.46) routes through L9 Feature 10769 "people, categories of people, or sports" → L19 → L21 → L25, providing a secondary clean excitatory contribution.

Convergent or ambiguous? Highly convergent. p=0.836 is the highest probability in the dataset. The circuit has two independent excitatory paths (Paths 2 and 5), multiple middle-layer features that independently encode the authorship concept, and a clean 2-hop shortcut from embedding to logit. The "person who writes books = author" concept is encoded redundantly across the network — early (book-writing context), middle (writing-project description), and late (creative-endeavors-and-authors) — producing a confident, well-supported prediction.

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4. Discussion

4.1 The Shared Linguistic Circuit: Recurring Feature Architecture

The 312 recurring features identified at the 50% threshold reveal a circuit that is predominantly populated by generic text-format and register features rather than linguistic-domain-specific detectors. Layer 0 accounts for 8 of the top 15 recurring features, uniformly low-activation nodes that encode positional and token-level context (e.g., Feature 13: "no matter how," node_id=0_13_4; Feature 6: "references to current study/research," node_id=0_6_1). These suggest that the sentence-boundary token `<bos>` systematically activates a shared set of context-anchoring features that fire regardless of the specific metalinguistic operation being performed.

Layer 6, Feature 2586668 (node_id=6_2267_3) is the most universally recurring feature (37 appearances / 10 graphs), labelled "*words that appear in programming code, legal jargon, or scientific texts*." Its cross-prompt universality reflects the fact that all 10 prompts share the formal metalinguistic template ("The X of Y is" / "A X of Y is called an"), which strongly resembles the register of technical/definitional writing. This feature is a template-format detector, not a semantic content detector.

The labeled top-15 features cluster into three functional groups consistent with the prompt-driven data flow:

1. **Template/register anchors (Layers 0–6):** Features 13, 6, 5, 17 (L0) and Feature 2586668 (L6) fire on the formal structure of the prompts.
2. **Morphological/lexical processing hubs (Layers 6–15):** Feature 12241 at Layer 11 ("name origins and meanings," recurring in antonym, plural, and tense prompts) and Feature 10933 at Layer 11 ("letters L, H, a at text beginning," very high activation in multiple prompts) serve as morphological transformation mediators.
3. **Output-selection features (Layers 15–25):** Feature 805 at Layer 15 ("proper nouns with preference for locations/nationalities") and Feature 3205 at Layer 3 (recurring 25 times) contribute to the answer-token selection stage.

4.2 Dominant Causal Paths and the Embedding Shortcut

The most consistent finding across all 10 prompts is the presence of a single dominant 1–2 hop excitatory path from the embedding aggregation node directly to the logit. In 8 of 10 prompts this path carries 85–99% of the total excitatory weight, with edge weights ranging from 3.63 ("atlas") to 14.14 ("mice"). This architecture implies that Gemma-2-2B largely resolves metalinguistic queries at the embedding stage — the SAE features at the final embedding position aggregate the answer signal before it is even passed through the late transformer layers. The late-layer features visible in the band tables are real circuit components but play supporting roles, fine-tuning competition among near-competitor tokens rather than generating the primary logit signal.

The two prompts with the weakest embedding-path dominance are also the two with the lowest prediction confidence: "written" (p=0.344, all paths mixed) and "swam" (p=0.496, five competing paths). In both cases the circuit lacks a clean single-origin excitatory anchor, suggesting that irregular past-tense computation is more distributed and contentious than irregular plural or synonym completion.

4.3 The Layer 24 Inhibitory Bottleneck

A striking architectural pattern is the convergence of causal paths through an unlabeled Layer 24 node that carries a large negative final edge to the logit. This bottleneck appears in Prompts 8 ("written," node_id=24_1633_6, edge=-3.56), 9 ("atlas," node_id=24_11962_7, edge=-6.89), and 10 ("author," node_id=24_1633_8, edge=-2.56). The recurrence of Layer 24 Feature 1633 specifically in both Prompt 8 and Prompt 10 suggests this is a systematic competitor-suppression mechanism: by routing activations through a strongly negative final gate, the circuit simultaneously eliminates multiple competitor tokens (e.g., "wrote," "write" for Prompt 8; "writer," "novelist" for Prompt 10) leaving the one correct answer as the least suppressed output. This negative-gate architecture at Layer 24 may be a general property of the model's factual and lexical completion mechanism, not specific to linguistic reasoning.

4.4 The Edge Neighbourhood Analysis: Top Recurring Features

The top 2 recurring features by average influence — Layer 6, Feature 2586668 (avg_inf=0.6571) and Layer 0, Feature 13 (avg_inf=0.5924) — reveal complementary roles in the circuit's wiring. Feature 2586668's incoming edges originate predominantly from Layers 0–5 (token-level embedding features), while its outgoing edges project to Layers 8–14 (middle-layer processing hubs). This confirms its role as an early-to-middle signal relay. Feature 13's edges are extremely sparse (zero-activation node), suggesting it functions as a structural anchor that marks sentence position rather than carrying semantic content.

4.5 Surprising Findings

Feature repurposing for morphological computation. The "baseball terminology" feature (Layer 5, Feature 10444, node_id=5_10444_3) achieved the highest early-layer influence score in Prompt 7 (0.8002) for the prompt "The past tense of swim is." Baseball verbs are predominantly irregular monosyllables with vowel-change past tenses (hit, pitch, throw, swim, run), suggesting the model has a polysemantic feature that fires on this morphological phonological class, not on sport semantics. This is a clear case of a feature being co-opted for an unexpected computational role.

Format-override of semantic retrieval (Prompts 1–4). Four of the ten prompts — all antonym/synonym prompts — predicted the colon ":" rather than the expected synonym/antonym. The circuit analysis for those prompts (detailed in Results Sections 3.4.1–3.4.4) revealed a format-following circuit that overrides semantic retrieval: a very high-activation recurring feature (Feature 110446948: "code snippets and license agreements") fires strongly for the "X is:" template, pushing the model to complete with a formatting character rather than a word. This is not a linguistic reasoning failure per se but a register-competition failure: the model's format-completion circuit outcompetes the semantic-retrieval circuit when the prompt strongly resembles list/definition formatting.

Layer 11, Feature 12241 as a cross-paradigm morphological hub. Feature 12241 ("name origins and meanings") recurred in both the irregular plural prompts (Prompt 5: "child", act=49.98; Prompt 6: "mouse", act=17.87) and the tense-change prompts, always at high activation. Its label ("name origins and meanings") is formally about etymology, but its activation pattern suggests it encodes the broader class of words whose current form derives from historical sound changes — irregular plurals, vowel-mutation past tenses, and suppletive forms — making it a functional morphological history detector.

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5. Causal Validation via Feature Steering

To test whether the causal paths identified in the per-prompt analyses above are genuinely load-bearing — not merely high-attribution but functionally inert — we ran causal steering experiments on three linguistic circuits using the Neuronpedia Steering API. For each circuit, we extracted the backbone features from the per-prompt top-5 causal paths and applied four tests: (A) **Necessity** — suppress each backbone feature individually at strength −20 and check whether the model's first predicted token changes; (B) **Full backbone suppression** — suppress all backbone features simultaneously; (C) **Sufficiency** — boost the circuit's primary hub at strength +20 on an altered prompt to test whether the hub alone can induce the target prediction; (D) **Specificity** — suppress non-backbone features (high-activation nodes not on the top-5 causal paths) to confirm they are causally inert.

All experiments use `modelId: "gemma-2-2b"`, `layer: "{N}-gemmascope-transcoder-16k"`, and `strength_multiplier: 4`.

5.1 Necessity (Individual Feature Suppression)

hot→cold (p=0.566, 7 backbone features)

Feature	Layer	Index	Role in Causal Path	Steered 1st Token	Disrupted?
L11_systems	11	4301	Path 1 entry	cold	no
L18_relay	18	1296	Mid-late relay	cold	no

L20_hub	20	6666	Central convergence (paths 1,2,3,5)	cold	no
L25_output	**25**	**16258**	**Output driver on all paths**	**not**	**YES**
L7_comparison	7	4526	Comparisons/symmetry feature	cold	no
L12_relay	12	95	Mid relay	cold	no
L14_relay	14	1354	Late-mid relay	cold	no

L25/16258 is the sole necessary feature — it appears on all multi-hop causal paths as the final node before the logit. Suppressing it changes the prediction from "cold" to "not" — the model shifts from antonym retrieval to negation, a semantically coherent failure mode that reveals the competition between "produce the opposite meaning" and "negate the adjective" as strategies for completing "The opposite of hot is." The L20 convergence hub (on 4 of 5 paths) is not individually necessary because the direct embedding→logit path (path 4, wt=9.30) bypasses it entirely, consistent with the Discussion's finding that the dominant embedding→logit shortcut carries the majority of the output signal.

maps→atlas (p=0.825, 7 backbone features)

Feature	Layer	Index	Role in Causal Path	Steered 1st Token	Disrupted?
L18_gateway	18	10679	Central gateway (edge +22.62)	atlas	no
L19_relay	19	8729	Post-gateway relay	atlas	no
L24_suppressor	**24**	**11962**	**Competitor suppression (edge −6.89)**	**"**	**YES**
L25_gate	25	8267	Final gate to logit	atlas	no
L10_references	10	6804	References feature (act=38.22)	atlas	no
L21_visual_info	21	3848	Visual representations	atlas	no
L16_word_origins	16	16329	Word origins/etymology	atlas	no

L24/11962 (competitor suppression bottleneck) is the sole necessary feature. Section 4.3 hypothesized that Layer 24 functions as a "systematic competitor-suppression gate" based on the recurrence of strongly negative final edges at this layer. This steering result causally confirms that hypothesis: suppressing the L24 gate changes "atlas" to a quotation mark — the model loses the ability to select among competing completions and falls through to punctuation. The L18 gateway (highest edge weight, +22.62) is not individually necessary because the direct embedding→logit path (path 3, wt=3.63) provides a bypass.

author (p=0.836, 9 backbone features)

Feature	Layer	Index	Role in Causal Path	Steered 1st Token	Disrupted?
L24_bottleneck	24	1633	Inhibitory bottleneck	author	no

L25_output	**25**	**14935**	**Output driver**	**(markup)**	**YES**
L22_gateway	22	7158	Excitatory gateway	author	no
L18_writing_projects	18	1943	Writing projects feature	author	no
L19_relay	19	7101	Relay	author	no
L21_relay	21	13124	Late relay	author	no
L25_amplifier	25	4717	Shared amplifier	author	no
L22_literature	22	9109	Literature/writing	author	no
L24_creative_works	24	1735	Creative endeavors/authors	author	no

L25/14935 is the sole necessary feature. The L24 bottleneck (Feature 1633) — the same feature identified in Section 4.3 as the Layer 24 inhibitory bottleneck shared with the "written" prompt — is not individually necessary here. The L22 excitatory gateway (direct 2-hop path, wt=66) provides sufficient redundancy. This suggests that while L24/1633 is an important architectural motif in the linguistic circuit, the author prompt's high confidence (p=0.836) routes the dominant signal through the excitatory L22 gateway rather than relying on the inhibitory suppression path.

5.2 Full Backbone Suppression

Circuit	Default Completion	Steered Completion	Disrupted?
hot→cold	...cold. The opposite	...not no matter no matter	**YES**
maps→atlas	...atlas. A book	...a a (whitespace)	**YES**
author	...author. A person	...<strong><strong><strong>...	**YES**

All three linguistic circuits are fully disrupted by simultaneous suppression of all backbone features. The failure modes are revealing:

- **hot→cold** degrades to "not no matter no matter" — the model enters a negation loop, consistent with the "not" failure mode seen when suppressing only L25/16258.
- **maps→atlas** collapses to article repetition, losing all lexical content.
- **author** produces HTML markup tags — the model falls through to a completely different register, suggesting the linguistic reasoning circuit and the formatting circuit compete for the same output slot.

5.3 Sufficiency (Hub Boost on Altered Prompts)

Circuit	Hub Boosted	Altered Prompt	Baseline → Boosted	Target Induced?
hot→cold	L11/4301	"The opposite of fast is"	slow → slow	no
maps→atlas	L18/10679	"A collection of maps is called an"	— → atlas	**YES**

maps→atlas	L18/10679	"A book of recipes is called a"	— → map of the world	no
author	L24/1633	"Someone who writes novels is called an"	— → author	**YES**
author	L24/1633	"A person who paints pictures is called an"	— → art(ist)	no

Sufficiency succeeds when the altered prompt retains the target semantic domain:

- The maps→atlas hub (L18/10679) induces "atlas" on "A collection of maps" — a near-paraphrase of the original prompt — confirming this feature specifically encodes the maps→atlas lexical mapping. On the cross-domain prompt ("A book of recipes"), it produces "map of the world" — the hub activates its map-domain association but cannot override the recipe context to produce "atlas."
- The author hub (L24/1633) induces "author" on "Someone who writes novels" but only "art(ist)" on "A person who paints pictures." The hub carries a "creative professional" concept that is necessary but not sufficient for the specific "author" prediction without writing-domain context.
- The hot→cold hub (L11/4301) fails to override "slow" on the "fast" antonym prompt, consistent with its role as an early-layer entry point rather than a domain-specific output driver.

5.4 Specificity (Non-Backbone Feature Suppression)

Circuit	Non-Backbone Feature	Neuronpedia Label	Steered 1st Token	Disrupted?
hot→cold	L2/8212	"compare/contrast"	cold	no
hot→cold	L0/902	"the word is"	cold	no
maps→atlas	L8/1494	"coding/GUI"	atlas	no
maps→atlas	L8/13791	"web dev/URLs"	atlas	no
author	**L10/1037**	**"capitalized words"**	**artist**	**YES**
author	L9/10769	"people/categories"	author	no

5 of 6 non-backbone features pass the specificity test. The sole exception is in the author circuit: suppressing L10/1037 ("capitalized words") changes "author" to "artist." This feature was classified as non-backbone because it does not appear on the top-5 causal paths, yet it causally contributes to disambiguating between semantically close completions ("author" vs. "artist" — both creative professionals, both likely capitalized in certain contexts). This false negative reveals a limitation of top-5 causal path tracing: features that contribute to fine-grained token discrimination may be excluded from the backbone if their causal influence is distributed across many weak paths rather than concentrated in one strong path.

5.5 Summary

Circuit	Confidence	Necessary Features	Full Suppress	Sufficiency	Specificity
hot→cold	0.566	1/7 (L25/16258)	DISRUPTED	0/1	PASS (0/2)

maps→atlas	0.825	1/7 (L24/11962)	DISRUPTED	1/2	PASS (0/2)
author	0.836	1/9 (L25/14935)	DISRUPTED	1/2	PARTIAL (1/2)

The causal validation confirms three key architectural claims from the Discussion:

1. **The Layer 24 inhibitory bottleneck is causally real** (Section 4.3). The maps→atlas circuit's L24/11962 suppression gate is the sole individually necessary feature — suppressing it alone is sufficient to disrupt the prediction. This validates the hypothesis that the Layer 24 negative-gate architecture is a functional competitor-suppression mechanism, not merely a correlate of high attribution weight.
2. **The embedding→logit shortcut provides redundancy** (Section 4.2). Most backbone features on multi-hop paths are individually dispensable because the direct 1–2 hop embedding→logit path (weights 3.63–14.14) carries the dominant signal independently. Only the final L24–L25 bottleneck features — which sit downstream of the shortcut — are uniquely necessary.
3. **Individual necessity is sparse** (mean 1.0 necessary features out of 7.7 tested per circuit), consistent with the highly parallel causal path structure documented in the per-prompt analyses.

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6. Limitations

Single model, single SAE. All analyses use Gemma-2-2B with the Gemmascope 16K transcoder SAE. Different SAE sizes or a different model family may produce different feature decompositions, recurring components, and path structures. The findings should not be assumed to generalise across model architectures.

Sparse causal path tracing. The `trace_causal_paths()` function traces only the strongest edge-weight chains (greedy forward/backward search). It will miss paths that are collectively important but individually weak. The 5-path limit per prompt means that weaker but mechanistically significant paths may be systematically excluded.

Unlabeled bottleneck nodes. Several of the most mechanistically critical nodes in the causal paths — including the Layer 24 bottleneck nodes and several Layer 25 intermediaries — lack Neuronpedia labels (returning "no label" from the feature explanation API). The functional roles attributed to these nodes in this paper are inferred from context, not from direct feature activation evidence.

Token competition table sparsity. The token competition tables for all 10 prompts were empty due to the absence of logit-ranked feature data in the raw graph JSON. As a result, the competitor-token suppression analysis relies entirely on the causal path classifications (excitatory/inhibitory/mixed) rather than on direct comparison of logit scores for the top-k candidate tokens.

10-prompt sample size. With only 10 graphs, the 50% threshold requires a feature to appear in at least 5 graphs. This is sufficient to distinguish universal circuit components from prompt-specific noise, but the power to detect sub-category-specific components (e.g., features specific to plural morphology vs. past-tense morphology) is limited. A larger prompt set with 20–50 examples per sub-category would allow finer-grained circuit dissection.

Polysemantic feature interpretation. Many of the identified recurring features are polysemantic (e.g., Feature 10199: "code snippets, family relations, and super sentai teams"; Feature 10444: "baseball terminology"). The functional roles assigned in the mechanistic narratives are the most plausible interpretations given the prompt context, but alternative interpretations cannot be excluded without targeted activation patching experiments.

Causal validation coverage. Steering experiments were run on 3 of 10 prompts (hot/cold, maps/atlas, author). The 4 antonym/synonym prompts that predicted ":" instead of the expected word, and the 3 morphological prompts (plurals, past tenses), were not causally validated. Extending validation to the format-override cases would test whether suppressing the format-completion circuit (L4/110446948) is sufficient to restore semantic retrieval.

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7. Conclusion

This paper presents the first detailed mechanistic analysis of the linguistic reasoning circuit in Gemma-2-2B across four metalinguistic sub-tasks. Our main findings are:

1. **A 312-feature shared circuit** underlies all 10 linguistic prompts at the 50% recurrence threshold, dominated by template-format and register-detection features rather than task-specific linguistic detectors. Layer 6, Feature 2586668 is the most universal component, present in every prompt as a formal-register template sensor.
2. **A dominant embedding→logit shortcut** (1–2 hops, weights 3.63–14.14) carries the majority of the output signal in 8/10 prompts, suggesting that metalinguistic completions are largely resolved at the embedding aggregation stage. Prediction confidence is inversely correlated with morphological irregularity and the availability of a clean excitatory embedding path.
3. **A Layer 24 inhibitory bottleneck** (Feature 1633, node_id=24_1633) functions as a systematic competitor-suppression gate in at least three prompts, concentrating multiple mixed-polarity paths through a strongly negative final edge before the logit. This architecture allows simultaneous elimination of multiple near-competitors without requiring separate inhibitory features for each.
4. **Feature repurposing** is pervasive: "baseball terminology" detects vowel-change monosyllables, "name origins and meanings" detects morphological history, and "code snippets and license agreements" activates for formal definitional templates. The model's linguistic circuit is built from general-purpose polysemantic features co-opted for specific sub-computations.
5. **Format-override failures** occur when the prompt register (formal, list-like) is sufficiently strong to activate the format-completion circuit instead of the semantic-retrieval circuit, as seen in 4/10 prompts that predicted ":" instead of the target lexical item.

These findings support the emerging view that language model circuits are opportunistic assemblies of polysemantic features with distributed, context-dependent roles, rather than dedicated modules for specific linguistic operations.